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TOPICAL REVIEW

A Review of Applicable Technologies, Routing Protocols, Requirements, and Architecture for Disaster Area Networks

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ABSTRACT The swift progress of various cutting-edge network topologies, especially Fifth-Generation Networks (5G), Sixth-Generation Networks (6G), Unmanned Aerial Vehicles (UAVs), and satellite technologies, presents a promising opportunity to guarantee reliable communication and offer consistent and secure emergency services in times of calamities. Fuelled by the rapid evolution of network architectures, this study provides a comprehensive overview of the emerging network topologies, technologies, routing protocols, and difficulties associated with disaster area networks (DANs). This investigation seeks to examine the integration of various DAN topologies, including aerial, terrestrial, and maritime networks, to enhance communication during a disaster. Subsequently, we assess numerous network performance metrics, such as delay, overhead, throughput, energy consumption, packet ratio, and security. The review is concluded by illuminating recent research trends and open issues in the domain of emergency communication. Ultimately, this review serves as a valuable resource for researchers, network engineers, and policymakers in the domain of DANs, offering a thorough basis for informed decision-making to streamline network deployments and preserve lives.

INDEX TERMS Ad hoc network, artificial intelligence, disaster area network (DAN), deep learning, emergency communications, geological disaster, Internet of Things, routing protocol.

I. INTRODUCTION

Recognizing the profound impact of natural and anthropogenic catastrophes, preserving communications and infrastructure is essential to protecting lives and property during emergencies. Consequently, launching a well-structured national disaster management plan involving strategies and procedures is vital to fulfilling humanitarian aims. A disaster management chain encompasses preparedness, response, mitigation, and recovery activities [1], which ultimately formulate effective pre- and post-disaster strategies to deal with large-scale hazards and save lives. Developing a comprehensive disaster management strategy requires guidelines for recovering communications infrastructure and deploying networks, together with procedures that delineate network design and parameters [2]. Truthfully, many nations are vulnerable to natural and human-caused disasters, including floods, mudslides, terrorist attacks, and industrial mishaps [3]. Malaysia is regrettably susceptible to flash floods due to severe meteorological conditions, with figures indicating an average of 143 floods annually. Flash floods occurred across Malaysia in December 2021, forcing 14,459 people to seek shelter and causing 48 casualties [4].

One of the paramount challenges following a disaster arises from maintaining communication among numerous entities, including the federal and local governments, hospitals,

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patrols, rescue teams, volunteers, and victims. A communication failure following disasters impedes rescue operations. The disruption of communication hinders interaction between the victim and rescue personnel, restricting the ability to convey information and ascertain the victim's location or medical condition. Hence, the rapid deployment of untraditional network systems is imperative to retaining communication during emergencies [5].

In contrast to traditional cellular networks, ad-hoc networks offer outshining characteristics that can be exploited in disaster situations. The characteristics of an ad-hoc network, which is a wireless network, include self-configuration, temporal deployability, and the absence of infrastructure. Therefore, the above traits are essential in emergency intervention, aid processes, and catastrophe relief [6].

In the same context, effectively handling disasters necessitates the integration of interdisciplinary physical and cyberspace components. Stated differently, adopting cutting-edge technology is essential for successful disaster management, particularly for swiftly deploying resilient communication systems in affected regions. In the event of a disaster, the integration of physical and data systems encounters numerous challenges. Therefore, a viable solution to guarantee optimal performance and seamless maintenance of vital services involves implementing a resilient network architecture that addresses and adapts to unforeseen obstacles [7].

Moreover, artificial intelligence (AI) and the Internet of Things (IoT) provide vital methods and technologies that can be applied to disaster management. Artificial intelligence offers various techniques that have a beneficial impact on rescuing victims. AI techniques comprise machine learning, network services, and analysis [8]. In parallel, IoT presents a feeder for AI algorithms to make comprehensive pre- and post-disaster predictions. IoT architecture and applications enable various devices to connect and share information, which leads to an increase in the amount of valuable information used in disaster management [9].

Managing disasters, whether foreseen or unforeseen, demands the confluence of efforts to reduce serious harm to individuals and property. Therefore, enhancing disaster management models entails the development of early warning systems, risk estimation plans, mitigation procedures, and recovery techniques applicable in disaster-prone regions. Several tools and approaches, such as Geographical Information Systems (GIS), spatial data management, sensor networks and the Internet of Things, big data analysis, and cloud computing, have been integrated to enhance decision-making and resource distribution in catastrophe situations [10]. In addition, effectively managing disaster area networks is essential for risk reduction during calamities, encompassing the deployment, management, and integration of network topologies. However, the infrastructure damage in the aftermath of the disaster may lead to the failure of the terrestrial network. Hence, the amalgamation of various network topologies, known as the space-air-ground integrated network (SAGIN), offers a feasible solution to reliable emergency communication. SAGIN architecture is divided into three distinct categories: space, air, and ground, including various network topologies such as satellite, unmanned aerial vehicles or balloons, and vehicular ad-hoc networks, respectively [11].

With the rapid development of intelligent air-ground integrated network architecture, UAVs represent a vital component that connects high-altitude and terrestrial networks. The hierarchical topology, like Lasagna, is designed to leverage the advancement of various UAVs, software-defined networking (SDN), and AI technologies. Therefore, a three-layer architecture comprises ground, low-altitude, and high-altitude platforms deployed to support rescue operations and intelligently manage communication using SDN and AI techniques during disasters. Deploying an integrated network poses various obstacles, including compatibility between three-layer components and issues related to security and privacy [12].

Despite extensive research on DANs, there is no unified framework that evaluates and compares topologies, technologies, protocols, and requirements in the context of cutting-edge emergency communication. The absence of a systematic analysis poses difficulties for researchers, network engineers, and policymakers to identify the most effective approaches for maintaining reliable communication and facilitating emergency services during calamity. Additionally, the integration of multiple advanced network topologies, encompassing aerial, terrestrial, and maritime networks, introduces communication complexities, such as delay, overhead, throughput, energy consumption, packet ratio, and security. Accordingly, a comprehensive analysis of the integration of cutting-edge technologies is necessary to stimulate researchers to continue exploring in the same field. This, in turn, facilitates the incorporation of cutting-edge network designs, technologies, and protocols, providing innovative solutions for efficient disaster management.

This review thoroughly examines cutting-edge DANs, highlighting relevant technologies, routing protocols, traits, requirements, and architectures. We systematically define current DAN topologies and their features. In addition, we evaluate existing DAN implementations, focusing on their effectiveness in guaranteeing emergency communication. Furthermore, we discuss the role of existing protocols, AI mechanisms, and IoT technology in enhancing emergency services. Through this review, we aim to provide valuable insights for researchers and professionals in the realm of DANs while identifying key areas for further investigation and development.

The following outlines the contributions of this survey:

- Present a comprehensive literature review of state-of-the-art disaster network topologies and technologies, along with key protocols, to facilitate the design and deployment of an efficient disaster network.

FIGURE 1. Ad hoc network types, subclasses, characteristics, and applicable scenarios.

- Examine the prospects of integrating various network architectures in disaster zones and conduct an in-depth investigation of existing solutions for disaster management systems.
- Highlight the requirements of disaster area networks to operate emergency-efficient services, such as resilience, reliability, scalability, and security.
- Discuss the challenges and propose novel research avenues for enhancing the deployment of networks and ensuring emergency services in disaster-stricken areas.

This review is structured as follows: Section II presents a concise overview of DAN's architectures, characteristics, and implementation challenges. Section III examines the existing literature and novel disaster area network topologies in the domain of DANs. In Section IV, we delineate the research framework and the methodology for the literature review. Additionally, Section V comprehensively outlines the requirements of DANs and offers an assessment of modern network topologies and routing protocols. Section VI discusses the challenges and future directions for countering the research-related concerns. Finally, Section VII concisely concludes this study.

II. BACKGROUND

Calamities lead to widespread infrastructure collapse and large-scale power blackouts, while deployed disaster area networks (DANs) constitute a fundamental element for protecting lives following a disaster. Subsequently, the deployed DANs ought to possess certain traits, including being rapidly deployable, reliable, efficient, and stable, to mitigate the side effects and unforeseen risks resulting from disaster [13]. This section, however, outlines the DAN topologies currently in use and their features.

A. AD HOC NETWORKS

In the occurrence of a crisis, device-to-device communication is essential for relief efforts. Therefore, deploying networks with unique traits is required. The ad hoc network, a wireless network, represents a promising and evolved network that offers a wide variety of swift response topologies and adaptable features. Ad hoc networks expanded into different topologies, including wireless mesh networks (WMN), mobile ad hoc networks (MANET), accompanied by their respective subclasses, and wireless sensor networks (WSN). Furthermore, ad hoc networks rely on a variety of traits, such as dynamic topology, mobility, reliability, density, computational power, etc. [14]. Figure 1 illustrates ad hoc network types, their respective subclasses, associated characteristics, and applicable scenarios.

1) WIRELESS MESH NETWORK (WMN)

Leveraging the characteristics of the wireless mesh network, a multi-hop wireless network, topology in several critical scenarios, such as defensive communication and disaster mitigation efforts, offers a promising solution. While processing sensitive information, including video, data, and VoIP, presents a fundamental dilemma, WMN network topology enables uninterrupted communication in the case of potential router failures. Furthermore, the decentralized topology of WMS, with many switches, enables valuable functions such as communication between nodes, scalability, and connectivity with minimal infrastructure requirements in an emergency [15], [16].

Permanent wireless mesh routers (WMR) constitute the fundamental components of the WMN's mesh topology that establishes wireless communication via the internet. However, the main duty of WMRs is to build connections between nodes within a hop destination using the proper routing protocol, hence creating an internet link. Furthermore, integrating WMN with a variety of ad hoc network architectures, including mobile ad hoc networks and wireless sensor networks, is the ideal advancement for WMN [17]. Figure 2 illustrates the general topology of WMN integrated with different ad hoc networks.

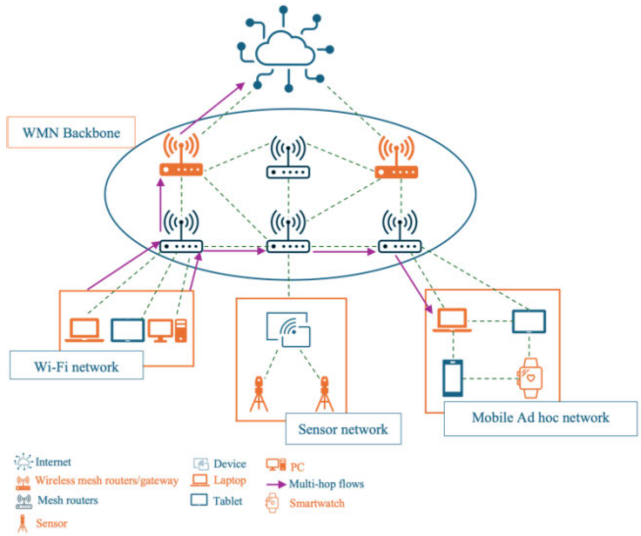

FIGURE 2. **The general topology of WMN.**

2) MOBILE AD HOC NETWORK (MANET)

In the context of mobile ad hoc networks (MANET), numerous significant traits encompass dynamic topology, multi-hop, absence of infrastructure, and simplicity of deployment. Moreover, the autonomous, expandable, and wireless link features provide further benefits to managing communication in challenging circumstances such as hurricanes, earthquakes, and tsunamis. In addition, individual nodes can establish a direct link without infrastructure, while routing protocols, such as reactive (on-demand) routing protocols, are designed to facilitate the data flow between nodes [18].

Another survey [19] concentrated on the primary constraints in establishing MANET. In particular, the authors focused on battery power management, which directly impacts the routing overhead and packet delivery ratio. Nevertheless, despite the previously mentioned obstacles, MANET persists in surpassing other network topologies due to the number of interconnected nodes and the absence of additional infrastructure. Figure 3 exhibits the MANET structure diagram.

Simultaneously, enhancing mobile communication demands the integration of several network topologies, including spatial, aerial, terrestrial, and maritime networks [20]. Therefore, this review explores the subclass of MANET topologies, which are detailed below:

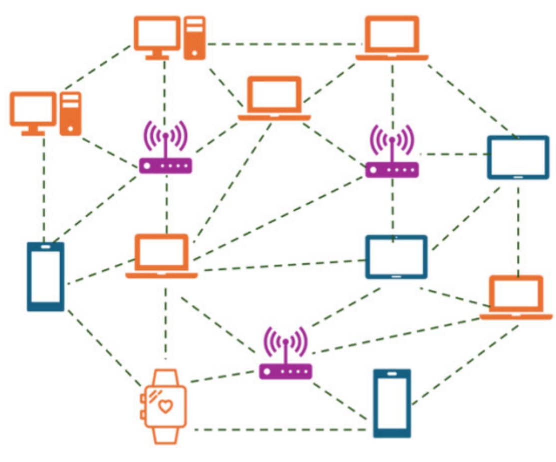

FIGURE 3. **The MANET structure diagram.**

1) **Vehicular Ad Hoc Network (VANET):** a wireless network, is a subset of MANET. The VANET topology consists of roadside unit (RSU) and vehicles equipped with onboard units (OBUs) that enable various types of communication. However, there are three types of communication over VANET: vehicle-to-vehicle (V2V), vehicle-to-roadside (V2R), and vehicle-to-infrastructure (V2I). Figure 4 illustrates the V2V, V2R, and V2I communication. Moreover, the system's communication range relies on the technologies

implemented in VANET for both the security and efficiency of linked vehicles. Within the VANET realm, the prevailing technologies are dedicated short-range communications (DSRCs) and long-term evolution vehicle-to-everything (LTE-V2X). The environment of VANET presents several challenges, including multimedia transmission, bandwidth limitation, and resource availability, which artificial intelligence can effectively address [21]. Real-time applications are crucial for maintaining sustainable contact and communication among V2V, V2R, and V2I, while the updated data encompasses news, weather, videos, and entertaining programs. Applications on VANET typically fall into the categories of safety, traffic, entertainment, and comfort. However, those applications possess different problems and difficulties including risk analysis, data verification, prioritisation and packet congestion control [22].

2) **Fly Ad Hoc Network (FANET):** Drones, also known as unmanned aerial vehicles (UAVs), constitute the Fly Ad Hoc Network (FANET), an extremely promising avenue for swift emergency responses in the aftermath of catastrophic events. The aim behind deploying FANET is to perform numerous crucial tasks, particularly disaster monitoring and management. Various drone features, such as size and weight, altitude, range, application, and flying mechanism, pose significant obstacles to FANET deployment. In addition, FANET's topology factors, such as power consumption, frequency band, radio propagation model, dynamic network architecture, node mobility, and node density, present major issues [23]. Regarding the previously mentioned issues, authors in [24] provided a comprehensive overview of relevant research on AI and ML-based solutions for different challenges in FANET-based networks. Meanwhile, incorporating several techniques is required to improve security, resource allocation, and precision UAV positioning in FANET. Furthermore, IoT, notably LoRa wireless technology, is essential in handling the FANET challenges, which demand low-power and long-range communications technology. FANETs can adopt various architectures, including single, multi-layer, and multi-group topologies. Leveraging these topological traits allows for efficient communication in times of emergency [25]. Figure 5 depicts the single, multi-layer, and multi-group topologies in FANET.

3) **Sea Ad Hoc Network (SANET):** also known as maritime wireless communication, is a network topology specifically intended for carrying out a range of maritime activities. The necessity to exclude human involvement, however, stems from the significant hazards associated with maritime activities. Therefore, cutting-edge technology, such as unmanned marine vehicles (UMVs) that fall into the categories of unmanned underwater vehicles (UUVs) and unmanned surface vehicles (USVs), is essential for eradicating hazardous maritime operations and boosting emergency communication [26].

Furthermore, employing standard maritime rescue research systems, such as autonomous surface vessels, underwater vehicles, or aerial platforms, presents considerable difficulties. Fortunately, using contemporary technology, such as unmanned systems, boosts SANET's performance in terms of high efficiency, cost-effectiveness, and swift deployment [27]. Figure 6 illustrates SANET's topology and communication types.

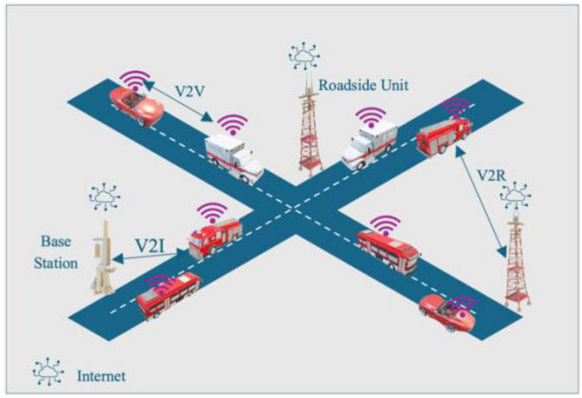

FIGURE 4. The V2V, V2R, and V2I communication.

3) WIRELESS SENSOR NETWORKS (WSN)

Presently, there is a widespread surge in research and queries regarding wireless sensor networks (WSNs) that leverage the Internet of Things (IoT). The importance of employing WSN stems from the technologies involved, which include wireless communication, an extensive spectrum of real-time applications, and a diverse range of sensors serving various purposes. Moreover, WSNs play an essential role in emergency communication during severe disaster scenarios, such as fire management. Despite this, the implementation of WSNs eliminates the limitations associated with other technologies encompassing wired systems, cameras, satellite systems, and Bluetooth. Numerous obstacles are present, including energy consumption, scalability, security, quality of service, network reliability, and data management. Consequently, the previous obstacles have prompted the proposal of several solutions that rely on machine learning algorithms and clustering methods [28], [29].

Furthermore, the limited number of simulators under assessment, the omission of important features, and the lack of explicit performance criteria raise challenges in selecting the optimal simulator for evaluating WSNs [30]. Figure 7 illustrates the WSN topology.

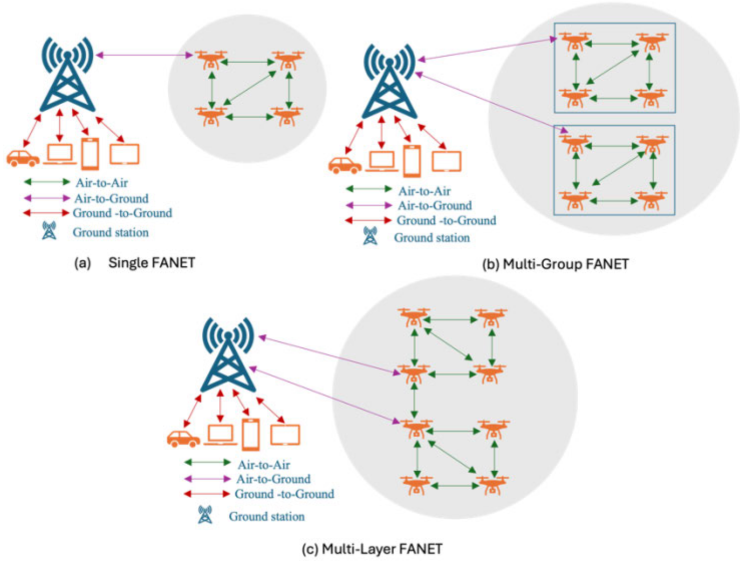

FIGURE 5. The FANET topologies.

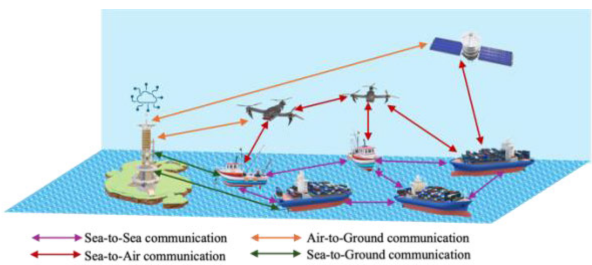

FIGURE 6. The SANET's topology.

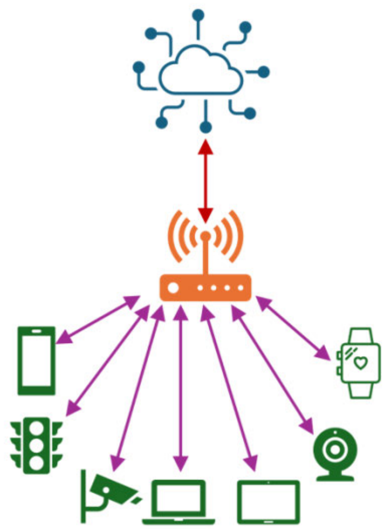

FIGURE 7. The WSN topology.

B. CELLULAR NETWORKS

Cellular networks, commonly known as mobile networks, offer the fundamental infrastructure and connectivity necessary for effective message exchange among a vast number of mobile devices across various geographical regions. While cellular networks make a significant contribution to improving service quality in disaster-affected areas, deployment of mobile networks poses notable challenges, such as scalability, throughput efficiency, overhead, etc., which require creative solutions to guarantee long-term viability and flexibility. However, the evolution of mobile networks, specifically 4G, 5G, and 6G, fosters the continued development of smart cities. This, in turn, facilitates the delivery of diverse services like emergency vehicle navigation, instantaneous data sharing, and traffic control [31].

The advancements in fifth-generation (5G)/beyond 5G (B5G)/six-generation (6G) technologies exhibit a remarkable capacity to detect, manage, and address forthcoming pandemics through the incorporation of the Internet of Medical Things (IoMT). The advancements in fifth-generation (5G)/beyond 5G (B5G)/six-generation (6G) technologies exhibit a remarkable capacity to detect, manage, and address forthcoming pandemics through the incorporation of the Internet of Medical Things (IoMT). Hence, several scholars concentrate on using significantly higher data rates, decreased latency, increased capacity, and a more efficient spectrum of next-generation technology for establishing networks [32].

Additionally, light is shed on the latest research trends in the 6G technology, which enables a terahertz communication system operating at T bits per second (Tbs) [33]. Nevertheless, 6G technology offers a wide range of key enhancements, including reduced latency, enhanced energy efficiency, increased data packet rate, greater connection density, and improved spectral efficiency. Thus, the previous major improvements of 6G enable the development of three-dimensional (3D) coverage communication networks across various environments, including underwater, terrestrial, aerial, and space, as well as digital twins, extended reality (AR), and virtual reality (VR) applications. Figure 8 illustrates the cellular network topology.

C. SATELLITE NETWORKS

Current research has primarily focused on satellite communication techniques for resilient communication in rural areas or disaster-affected regions. Researchers are exploring ways to provide diverse services that include telecommunications, radio, broadcast, computing, and military applications. In addition, the deployment of a variety of satellites, comprising geosynchronous equatorial orbit (GEO) satellites, medium earth orbit (MEO) satellites, low earth orbit (LEO) satellites, and ground stations, is an effective approach for tackling emergency communication challenges. Despite this, a complex three-dimensional hierarchical network that includes the previous topologies not only enables effective communication but also poses additional obstacles such as administration, integration, and security complexities [34], [35]. Figure 9 depicts the satellite network architecture.

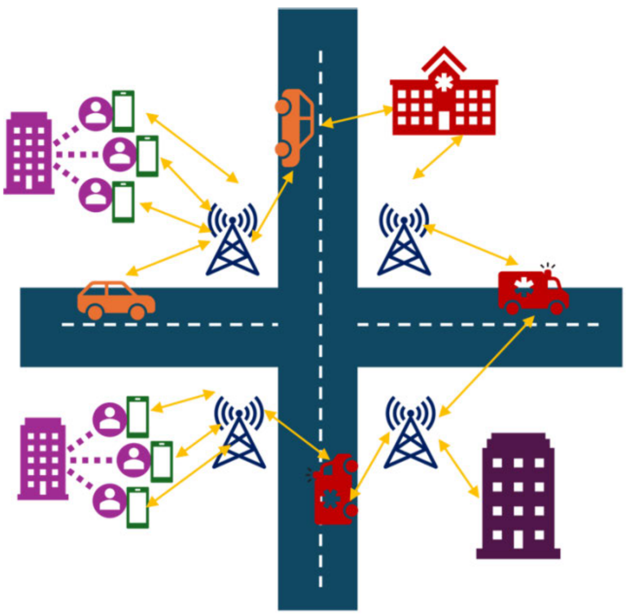

FIGURE 8. The cellular network topology.

III. RELATED WORKS

Numerous studies have been conducted to leverage the deployment of resilient and non-infrastructure networks during catastrophes to boost rescue operations. While integrating several network topologies, such as mobile ad-hoc networks (MANET), vehicular ad-hoc networks (VANET), and wireless sensor networks (WSN), is essential, this integration aims to enhance communication among rescuers, victims, and emergency authorities. Accordingly, various pieces of literature have been carried out on related topics. This section covers the literature on disaster area network technologies, routing protocols, and architecture.

The research in [36] offered an extensive survey of the pairing of 5G-and-beyond networks in multiple unmanned aerial vehicles (UAVs) to enhance network resilience, reliability, coverage, and cost. The authors discussed the collaboration of various emerging technologies and applications to present salient features for drones, 5G, and wireless networks. Additionally, this survey covered the fundamental aspects of exploiting a range of network topologies, the Internet of Things (IoT), and AI techniques to protect lives during floods, storms, tornadoes, and severe snowstorms. Despite the promising advancements in the cutting-edge technologies mentioned, several challenges emerge, including collision prevention, control latency, energy constraints, and security.

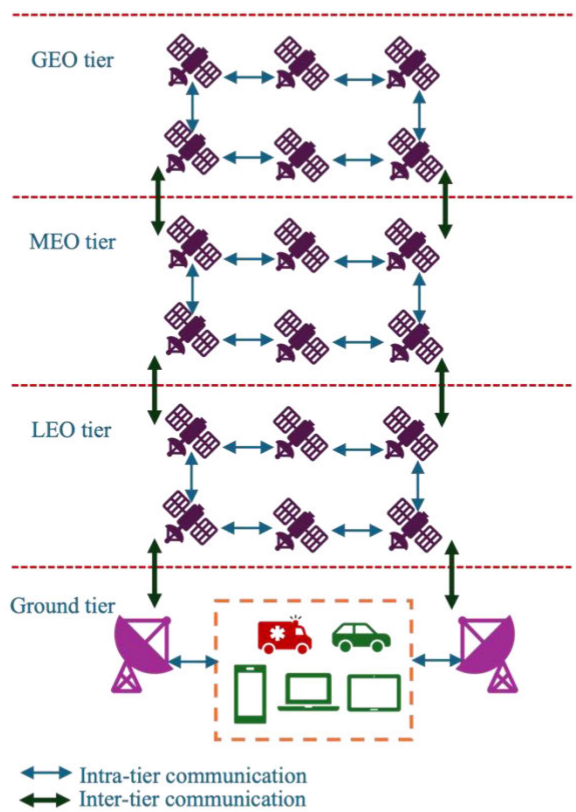

FIGURE 9. The satellite network architecture.

In [37], the authors paid more attention to artificial neural network (ANN) approaches for risk management in times of calamity, which are popularly used to improve resilience and rapid deployment. The study fully clarifies the reasons behind the supremacy of ANN-based approaches compared to other approaches relative to performance. While researchers highlighted the phases ranging from alleviation and planning to response and recovery, they also outlined the different types of catastrophes, including floods, wildfires, and others. Afterwards, a thorough analysis was carried out, demonstrating the effectiveness of employing ANNs to forecast and control the risk of flooding. Subsequently, they presented prospects for the future that encompassed the potential for developing hybrid ANN techniques for making disaster-related decisions, addressing the issue of data shortages, and incorporating the global positioning system (GPS).

The reference research [38] explored the capability to enhance the earthquake early warning systems (EEWS) by leveraging the IoT and cloud architecture. Furthermore, the survey examined the deployment of IoT networks for monitoring seismic waves and enabling cloud architecture to collect data, assess, and emit alarms. The authors additionally covered several novel techniques to boost disaster network deployment and scalability, including machine learning (ML) algorithms, distributed computing, and edge computing. Regarding future directions, the research suggested some methods to improve the EEWS, consisting of establishing precise and effective sensors, the fusion of artificial intelligence (AI) and ML, the standardization of communication protocols, and the use of free, open-source software.

The research in [39] provides a comprehensive synopsis of the significance of employing the sensor as a primary deployed component during catastrophe management. In conjunction with the types of deployed sensors, incorporating wireless sensor networks (WSN) and Internet of Things (IoT) techniques play a pivotal role in designing effective disaster management networks. A design that considers a few crisis management responsibilities, among them finding victims, lowering risk, and prompt tracking. Moreover, the research aims to effectively integrate emerging technologies to support deployable calamity sensors. The supported sensor technologies include computer vision, AI, auditory systems, robots, WSNs, and IoT, which integrate to enhance interoperability, mobility, real-time data, and scalability. However, the permanent sensors' spots are susceptible to damage under certain circumstances. In the future, the emphasis will be placed on deploying WSNs and UAVs to ensure real-time communication and convenient sensor mobility, respectively.

In [40], the authors placed greater emphasis on revolutionary sixth-generation (6G) technology to improve data throughput and reliability. While 6G technology will surface following the global deployment of fifth-generation (5G), the study investigates the need for 6G technology and its implications for vital sectors such as public protection and disaster relief (PPDR), autonomous vehicles, and smart transportation systems. In addition, the study explores the challenges and potential of combining 6G with contemporary technologies, including terahertz communication, artificial intelligence (AI), machine learning (ML), and quantum communication. Implementing 6G technology involves significant obstacles, including limited bandwidth, high investment costs, and security and privacy concerns that require thoughtful consideration.

Table 1 introduces a review of prior studies, accompanied by an analysis and comparison of contributions for the reader's better comprehension.

IV. RESEARCH METHODOLOGY

This section defines the research framework. Furthermore, this section outlines the process of searching for literature and its various classifications.

A. LITERATURE SEARCH

We initiated our literature review by studying three specific domains of disaster area networks: technologies, routing protocols, and architecture. We commenced our search using keywords such as Disaster Area Network (DAN), Ad Hoc Network, routing, emergency communications, geological disasters, artificial intelligence, deep learning, and IoT integration. Moreover, the digital libraries and research platforms used in our examination are Google Scholar, Springer Nature, IEEE Xplore, Multidisciplinary Digital Publishing Institute (MDPI), and Elsevier ScienceDirect. Initially, our search yielded over 300 articles with publishing years between 2019 and 2024. After examining the existing literature and research trends, we concentrated on relevant technologies, routing protocols, and architecture for disaster area networks. Hence, we enhanced our search for literature using keywords like ad hoc networks, emergency communications, wireless mesh networks, mobile ad hoc networks, vehicular ad hoc networks, fly ad hoc networks, sea ad hoc networks, wireless sensor networks, and cellular networks. The enhanced search has yielded more than 130 articles related to our research.

B. LITERATURE TAXONOMY

The authors used their own filtering choices to determine the ultimate selection of publications. During the first phase, we concentrated on recent review articles relevant to the research issue, particularly those published from 2019 to 2024. In the second phase, we identified research articles focusing on the most recent applicable technologies, routing protocols, and architectures in the domain of disaster area networks backed by trustworthy publishers like Elsevier ScienceDirect and Springer Nature. Our study comprises 31.4% review articles and 68.6% research articles.

The majority of our selection consists of publishing years, as illustrated in Table 2, from 2019 to 2024. Additionally, Table 2 demonstrates that the Google Scholar research platform comprises approximately 32.6% of the articles. On the contrary, IEEE Xplore and MDPI occupy the second and third spots, at around 27.21% and 20.4%, respectively. We have systematically selected the timeframe to examine and review the emergence and evolution of disaster area networks in the past few years. Therefore, reviews and research articles mentioned in our research show promising solutions in the domain of disaster management by leveraging different network topologies and technologies to face emerging global threats.

V. DISASTER AREA NETWORK: REQUIREMENT AND COMPARATIVE EVALUATION

Disaster area networks offer effective solutions by sustaining communication in regions with damaged or non-existent

TABLE 1. Overview of prior studies on disaster area networks.

Ref.	Research trajectory and contribution	Primary scope of research	Limitations/Gaps
[36]	Driving progress in UAV communication technologies to meet the emerging requirements.	UAV, Opportunistic routing (OR), Highest Velocity Opportunistic Routing (HVOR), 5G-and beyond, IoT, AI	Dedicated solely to integrating 5G and beyond networks with UAV systems.
[37]	Explored the efficacy of Artificial Neural Network (ANN) techniques in the mitigation and preparation of disasters.	ANN to predict and manage flood, Backpropagation Neural Networks (BPNN), Convolutional Neural Networks (CNN)	Exhibits limited capability in predicting, identifying, and classifying post-disaster activities. Constrained by challenges associated with big data.
[38]	A thorough evaluation of the taxonomy of the emerging Earthquake Early Warning System (EEWS) approach leveraging IoT and cloud technology.	IoT, cloud infrastructure, sensor network	Constrained by the scope of the sensor network, with limitations in guaranteeing reliability, accuracy and real-time data processing.
[39]	Evaluated the performance of different sets of sensors to design optimal systems aimed at mitigating disaster risk by integrating wireless networks for catastrophe detection and survivor tracking.	IoT, robots, WSNs	Narrow focus on WSNs and UAVs to ensure real-time communication.
[40]	Investigated the potential of the integration of 6G in public protection and disaster relief systems.	6G, terahertz communication, (AI), machine learning (ML), and quantum communication	High investment costs, along with challenges related to security and privacy
This Review	Exhibited an evaluation of the incorporation of various DAN topologies, including aerial, terrestrial, and maritime networks, alongside examining a range of emerging technologies and protocols, analysing their advantages, potential applications and technical hurdles.	5G, 6G, IoT, AI, Sensors, machine-learning, Ad Hoc Networks (WMN, MANET, WSN), Cellular Networks, Satellite Networks, Ad hoc On-demand Distance Vector Routing (AODV) protocol, the Destination-Sequenced Distance Vector (DSDV) routing protocol etc.	Not adaptable to low-cost alternatives and renewable energy technologies

infrastructure. Thus, these networks are engineered to operate in severe conditions, facilitating vital functions such as real-time data transmission and emergency services. Moreover, evaluating the performance of DAN is crucial to guarantee network resilience and reliability in the event of a calamity.

Metrics such as scalability, reliability, latency, throughput, and energy efficiency are essential for assessing the capability of these networks under different scenarios. This section delineates the fundamental requirements of DAN and presents a comparative evaluation of modern network topologies in disaster areas.

A. REQUIREMENT OF DISASTER AREA NETWORK

Network engineers design and develop a disaster area network to operate essential emergency functions under severe and unpredictable circumstances. The network must continue functioning despite significant disruptions or compromises to conventional communication infrastructure. Consequently, sustaining communication during emergencies necessitates paying attention to multiple requirements, including resilience, reliability, scalability, swift deployment, energy efficiency, and security. This section sheds light on the requirements that must be considered in the disaster network design process.

1) RESILIENCE AND RELIABILITY

Undoubtedly, a key requirement for guaranteeing recovery following a network failure is to increase the network's resilience. Thus, deploying a disaster recovery framework is imperative. The meticulously structured framework demands the implementation of a circular and iterative process, which includes ongoing preparedness and recovery planning, relief and short-term recovery, long-term recovery, and transition [41]. Integrating software-defined networks into a UAV enables the employment of a multipath routing method,

TABLE 2. The aggregation of review and research articles obtained through various criteria.

Publishing years				
2019-2020	2021-2022	2023	2024	
3	11	53	80	
Digital libraries and research platforms				
Elsevier ScienceDirect	IEEE Xplore	Springer Nature	Multidisciplinary Digital Publishing Institute (MDPI)	Google Scholar
18	40	11	30	48

which enhances the network's resilience, especially in emergencies. Furthermore, the mesh network topology, which often connects nodes via many pathways, facilitates the achievement of the same goal. Additionally, the implementation of swarm intelligence algorithms and the use of LoRa technology significantly boost network resiliency [42].

From another perspective, disaster area networks are susceptible to failure in extreme conditions due to the characteristics of network nodes, including wearable devices, mobiles, UAVs, and sensors. Thus, enhancing network reliability is fundamental to obtaining real-time data and ensuring fault tolerance. However, leveraging cutting-edge technology, such as deploying the Internet of Emergency Services (IoES) to collect real-time data, directly affected the reliability of the network [43]. In addition, cloud computing, particularly when combined with satellite remote sensing data and IoT, provides reliable earthquake emergency systems. The significance of cloud computing technology stems from its various features, including rapid data retrieval, extensive data storage, and communication capabilities [44]. Moreover, establishing a reliable monitoring system is essential for the post-disaster phase. The monitoring system comprises multiple components, including flying sensor networks (FSNets), cloud/fog computing, and AI algorithms integrated to satisfy the reliability requirement [45].

2) SCALABILITY

In the case of an emergency, a network needs to be sufficiently adaptive to cope with a fluctuating number of nodes, including victims, authorities, and on-site rescue teams. At this juncture, network engineers must evaluate the scalability requirement to accommodate the growing request for network resources. Therefore, using millimeter-wave (mmWave) communications, an ultra-fast and low-latency wireless system, presents a significant opportunity to exploit network resources effectively. The distinctive attributes of mmWave signals, such as an expanded bandwidth and exceptional data transfer rates, enhance network scalability in numerous scenarios [46].

Furthermore, deploying UAV-operated networks is necessary for obtaining real-time data, which is vital for decision-making in disaster-affected regions. Enhancing scalability necessitates the efficient management of UAV fleets, which demands the integration of a robust mathematical algorithm to determine the minimum number of aircraft. Hence, the deployment of a precisely designated quantity of UAVs facilitates optimal network resource allocation [47], [48]. Additionally, proactive and reactive protocols, each with distinct characteristics, enhance network efficiency by ensuring high data transfer rates, facilitating reliable communication, and reducing latency. The implementation of a position-based protocol in DAN, which enables the determination of next hops based on node locations, aims to reduce overhead and improve scalability [49].

3) COVERAGE

Expanding network coverage, particularly in rural areas and remote places, is crucial for delivering emergency services. However, drones continue to play a significant role in enhancing network coverage. Incorporating the reinforcement learning (RL) architecture and the Siamese-net neural network, which leverages a database of times and areas that drones previously visited, leads to enhanced coverage [50]. Moreover, the deployment of UAVs as mobile aerial base stations (MABSs) after the destruction of terrestrial infrastructure facilitates the preservation of communication and addresses coverage deficiencies. Leveraging the prior network architecture necessitates a combination of graph theory, a genetic algorithm (GA), and long-term evolution (LTE) technology, which together yield enhanced overall service and coverage through the affected regions [51].

Moreover, the damage to the fiber optic connectivity infrastructure poses an additional obstacle to rescue operations. Mitigating the lack of connectivity requires a hybrid combination of integrated access and backhaul (IAB) techniques and non-terrestrial networks (NTN), which is considered a promising method in such scenarios. The integration of IAB and NTN, which encompasses high- and low-altitude platforms (HAPs/LAPs) and low-earth orbit (LEO) satellites, addresses the issue of network coverage during crises [52]. In this context, the oscillating spider monkey optimization (OSMO) approach, which relies on a device-to-device

(D2D) communication mode, presents an alternative for cellular networks. The suggested approach ensures coverage by implementing cluster-based D2D communication [53]. Consequently, disaster network coverage can be classified into three distinct groups: local, regional, and global, based on the network topologies and techniques applied.

4) RAPID DEPLOYMENT

In the realm of disaster area networks, rapid network establishment is a vital requirement. Likewise, the ability to facilitate communication with minimal configuration and without relying on pre-existing infrastructure requires a seamless transition from a conventional network to an emergency network. Hence, addressing the preceding criteria may foster communication during critical situations. Satellite phone communication, which does not require terrestrial infrastructure, offers numerous features that facilitate and enhance rapid emergency communication. At the same time, this technology faces some obstacles, such as cost and power efficiency. However, incorporating advanced technologies, including AI algorithms, low-data-rate modulation schemes, low-bitrate voice codecs, and low-power encryption techniques, may enable a swift deployment at a reasonable cost [54].

Three-dimensional (3D) wireless networks offer another tangible solution for achieving rapid deployment through a combination of terrestrial, aerial, and satellite communication technologies. Notwithstanding the significance of deploying a 3D network framework, it still faces specific hurdles, including cybersecurity, cross-border coordination, and the physical layer [55]. Furthermore, the escalating use of UAVs contributes to the swift deployment of networks. Employing three-layer computing architectures, such as mobile edge computing (MEC) and vehicle fog computing (VFC), along with a UAV client layer, enhances connectivity and meets the rapid deployment requirements in DAN [56].

5) COST

The cost of deploying DAN, particularly in developing countries, presents a major barrier. From this perspective, deploying and integrating affordable hardware and software can effectively leverage DAN in communities with limited resources. Nevertheless, cloud computing, a low-cost computing method, allows users to receive emergency services on demand without necessitating prior hardware deployment or the installation of software on their devices [57]. Collaborative UAVs and edge intelligence provide several additional capabilities, including smart device connectivity and collecting and storing data in the cloud. Consequently, the edge intelligence platform of UAVs guarantees mobility, adaptability, and cost-effectiveness [58].

Geographical constraints do not limit satellite communication, allowing it to deliver emergency services in large areas with limited ground infrastructure. Although the deployment of satellites, UAVs, and terrestrial networks boosts DAN service, this hybrid architecture still confronts a significant challenge due to its cost [59]. Consequently, the incorporation of mini- and microsatellites presents a significant opportunity to deploy cost-effective networks. Furthermore, commercial firms offer services such as low-power wide-area networks (LPWANs) and spacecraft components. The preceding services enable the use of economical space technologies during a crisis [60].

Developing advanced approaches to boost the efficacy of the communication network in disaster areas (CNDA) during information propagation plays a crucial role in fulfilling network requirements, especially regarding optimal node density, communication range, and deployment cost. In this context, deploying the advanced multi-layer information dissemination model of CNDA (MMND) boosts network performance. The proposed model is segmented into three layers according to the information scope. The first layer determines the positioning of UAVs, while the second layer permits the sharing of critical information between UAVs and vehicle nodes. The third layer ensures the exchange of information pertinent to the specific rescue plan across vehicle and rescuer nodes. Thus, the proposed model guarantees optimal node density and communication range while steadily reducing the network deployment cost [61].

6) ENERGY EFFICIENCY

Disaster-hit regions are prone to the destruction of power sources, which frustrates efforts to extend battery life for mobile nodes. Improving energy efficiency in DANs requires enhanced techniques and protocols for data routing and transmission. Concurrently, the goal of applying a deep reinforcement learning-based algorithm is to develop an efficient multi-UAV framework. The proposed framework divides the disaster region into grid segments and enhances the UAV path, thereby alleviating issues related to UAV operations, such as battery lifetime and communication range [62].

Employing routing protocols is essential to guarantee energy efficiency in DAN. Thus, the recommended reactive protocol known as Distance and Energy-aware AODV (DEAODV) leads to the enhancement of network nodes' energy consumption. During the decision-making process, the DEAODV generates a routing matrix to identify the shortest trajectory and node energy, thereby extending the network's lifespan during a disaster [63]. Furthermore, the Disaster Scenario Optimal Link State Routing (DS-OLSR) protocol serves as a substitute for the Optimal Link State Routing (OLSR). Whereas implementing the OLSR causes rapid energy depletion, applying DS-OLSR yields energy savings. Therefore, DS-OLSR significantly enhances energy efficiency in emergency communication [64].

7) SECURITY

Security constitutes a fundamental requirement for DANs. The significance of meeting security requirements stems from the extensive accessibility of public networks, such as Wi-Fi and the Next G networks. Meanwhile, the enormous volume of sensitive data exchanged during emergencies presents an additional security challenge [65]. The dissemination of data during emergency communication, which contains historical, health, and personal information, is susceptible to cyberattacks. Wi-Fi and NextG networks meet security standards by using sophisticated protocols and techniques. However, security threats arise due to the constant communication between devices and the employment of emerging technologies. In consequence, DANs face several threats, including eavesdropping attacks, impersonation attacks, denial-of-service attacks, and message falsification attacks [66].

AI/ML modules provide precise methods of threat identification, especially in cloud emergency management systems. Incorporating intelligent cloud management facilitates cluster management in distributed areas that implement the Kubernetes cluster technique. The proposed approach offers essential security measures to detect possible and suspicious threats. In addition, it facilitates a prompt response to mitigate security issues in times of crisis [67]. An appropriately developed intrusion detection system (IDS) is essential for precisely monitoring and preserving sensitive data. IDS-enabled UAVs use a dynamic approach in an SDN-FANET to track and assess malicious activities. The IDS-enabled approach adopted a UAV resource-aware load balancing strategy to guarantee secure communication [68].

Enhancing security measures to improve the safety of emergency applications, particularly cloud computing apps, is crucial. Furthermore, preserving data in the cloud servers necessitates the deployment of a thorough protection solution [69]. However, employing the elliptic curve Diffie-Hellman (ECDH) algorithm contributes to the promotion of end-to-end data encryption. Authorised users can obtain data through the encryption mechanism, which involves exchanging keys based on ECDH. Implementation protection algorithms, such as digital signatures and MD5, may enhance the security of sensitive data during a disaster [70].

8) HIGH THROUGHPUT AND LOW LATENCY

Providing and sustaining effective emergency services, including video streaming, voice communication, and emergency warnings, is of paramount importance in DANs. Expanded emergency services require a massive information exchange; consequently, the emergency communication system should guarantee high throughput and low latency. Sophisticated emergency communication systems (ECSs) leverage hybrid network architectures, such as Wi-Fi, LPWAN, and SD-WAN, which facilitate VoIP services with low latency. Accordingly, the amalgamation of different network architectures enhances quality of service (QoS) and quality of experience (QoE) during catastrophes [71]. AI algorithms enhance the handling of vast amounts of data and elevated data rates by employing ML/DL techniques in threat surveillance systems. In turn, edge and cloud computing facilitate data manipulation and the use of computer resources, respectively. Therefore, implementing the prior computing techniques guarantees diminished latency, enhanced bandwidth, and the accurate assessment of large amounts of data in emergencies [72].

9) INTEROPERABILITY AND MOBILITY

Effective deployment of disaster area networks entails the adoption of numerous network topologies, including UAVs, Wi-Fi, SDN, LTE, satellites, etc., to facilitate emergency service. Moreover, DANs must incorporate various types of handwear and software to enable emergency assistance in heterogeneous environments and rapid mobility. Accordingly, the development of a sophisticated framework, such as a unified 3D network framework, aims to accommodate diverse mobility models and heterogeneous connectivity. The unified 3D network framework successfully combines space, air, and ground networks. The framework also uses advanced beamforming techniques, federated learning mechanisms, and energy-efficient computational offloading strategies to enhance network performance during disasters [73]. In turn, cross-platform software is vital for boosting user experience by offering integrations with different operating systems [74]. Furthermore, unregulated mobility poses major challenges for DANs due to the seamless movement of diverse devices, including wearables, mobiles, and drones. Nevertheless, enhancing mobility in UAV networks post-disaster necessitates the implementation of mathematical models, such as the long-distance path-loss model. The model simulates and forecasts fluctuations in signal strength across various distances in three-dimensional airspace [75].

B. A COMPARATIVE EVALUATION OF DISASTER AREA MODERN NETWORK TOPOLOGIES

Disaster Area Networks (DANs) are essential in preserving individuals' lives during severe events. Accordingly, several research reports exist on the benefits and drawbacks of employing the various topologies, technologies, and protocols. Thus, deploying a well-designed DAN required coupling multiple network architectures, methods, and routing algorithms to achieve a high QoS in times of crisis. The performance matrix is an indispensable component of the DAN evaluation approach, encompassing a range of indicators such as topology characteristics, implementation cost, reliability, packet latency, overhead, packet ratio, etc. In this context, this survey examines the previously mentioned publications to highlight their specific distinctions.

1) A COMPARATIVE EVALUATION OF MODERN NETWORK TOPOLOGIES AND TECHNOLOGIES

In the realm of DANs, several combinations of topologies and technologies, each possessing pros and cons, have been implemented to boost disaster relief operations. Despite that, a key feature of well-managed disasters is the integration of diverse architecture and techniques throughout the pre- and

TABLE 3. The WMN deployment characteristics.

Technologies / Techniques	Primary objective	Merits
IoT sensors and LoRa [76]	Enhancing data delivery, transmission range and scalability	Increasing communication range and delivery ratio
Bluetooth 5.0 technology [77]	Maintaining peer-to-peer (P2P) and mesh network communication	Self-healing capability and reroutes signals
Triangular and quadrangular mesh network designs based on connected safe set (CSS) [78]	Optimising router installation on specific mesh network nodes.	Allowed dual loads to endure faults and disruptions during disaster recovery
UAV and IoT devices [79]	Gathering data and sustaining communication	Developed an integer linear programming (ILP)-based model to enhance the UAV trajectories
Artificial intelligence of things [80]	Self-healing and network resilience	Developed a lightweight multi-task model to enable smart disaster preparedness

post-disaster stages, consequently enabling a swift response and protecting lives.

Numerous additional research fields exist that pertain to the implementation of WMN communications and networking to enhance data delivery, transmission range and scalability in critical times. Reference [76] proposed a WMN topology consisting of IoT sensors and LoRa, a long-range transmission technology and devices. The implemented network architecture shows an increasing communication range and delivery ratio in urban areas.

Authors in [77] evaluate the performance of deployed Bluetooth 5.0 technology to mitigate fire in disaster. Additionally, researchers have examined the value of the received signal strength indicator (RSSI) in peer-to-peer (P2P) and mesh network communication. According to the study, mesh communication coordination enables self-healing capability and reroutes signals to alternative pathways across extensive disaster areas. Furthermore, the design of WMN topology plays a crucial role in enhancing the installation of routes on specific mesh network nodes. In [78], authors introduced novel triangular and quadrangular mesh network designs based on connected safe set (CSS) nodes. These topologies allow dual loads to endure faults and disruptions during disaster recovery.

The incorporation of an unmanned aerial vehicle (UAV) and IoT devices to establish a mesh network is discussed in [79], exploring the ability to gather data and sustain communication in an infrastructure-free environment. Moreover, researchers developed an integer linear programming (ILP)-based model to enhance the UAV trajectories. The concept of fusing artificial intelligence of things (AIoT) for catastrophe management employing WMN architecture is elucidated in a notable study [80]. In addition to harnessing WMN features, including self-healing and network resilience, the authors developed a lightweight multi-task model to enable a smart disaster preparedness platform.

However, each WMN topology offers different advantages and considerations for appropriate communication during a disaster, and incorporating multiple technologies to design MWN poses a need for further investigation in this domain. Table 3 illustrates the WMN deployment characteristics mentioned in this section.

Owing to its numerous advantages, MANET is widely deployed to boost communication during natural disasters. Researchers integrated a sensor system for disaster communication and detection powered by a machine-learning algorithm into MANET [81]. In contrast, the study by [82] integrates affordable LoRa devices into MANET to facilitate long-range device-to-device communication via Bluetooth and Wi-Fi.

According to the MANET architecture, the quantity of connection nodes poses challenges in data transmission. An intelligent system design is necessary, considering the significant role of IoT and Artificial Bee Colony (ABC) algorithms in enhancing emergency communication. In [83], the study introduced an innovative intelligent system architecture in the MANET environment. Moreover, the MANET cashing mechanism plays a significant role in boosting network performance. With limited storage capacity, the management paradigm for data caching in the MANET is open to further refinement to ensure swift access to critical data. A holistic solution based on a hybrid caching algorithm is introduced in [84].

In [85], the researchers developed optimized resource allocation Using Tokens (RAUT) and queue management (QM) models for elastic (EL) and inelastic (IEL) traffic flows. The research aims to improve end-to-end delay (EED) and quality of service (QoS), which are critical factors in the communication services offered among MANET nodes.

Additionally, a significant obstacle in MANET is to enhance performance metrics, such as network throughput, latency, and energy consumption. The study by [86] proposed

TABLE 4. Various related technologies and techniques implemented in MANET.

Technologies / Techniques	Primary objective	Merits
Sensor and a machine-learning algorithm (GMMs) [81]	Boosting disaster communication and detection	Useful for observing various patterns in sensor data. The proposed control model is robust in terms of disaster forecasting and communication.
LoRa device, Bluetooth and Wi-Fi [82]	Enabling long-range device-to-device communication	Enabling the transmission and reception of long-distance messages. Implement chat software for iOS, Android, and laptop/desktop devices.
IoT and Artificial Bee Colony (ABC) [83]	Designing innovative intelligent system architecture	Enhancing emergency communication more specifically for data transmission
Hybrid cache placement and replacement approach [84]	Enhancing cache management strategies in MANETs	Improving cache hit ratio and average hop count.
RAUT and QM models [85]	Optimizing resource allocation and queue management	Enhancing the Quality of Service (QoS). Improvement of the data transmission rate for applications like VoIP.
Genetic algorithm, particle swarm optimization, and ant colony optimization [86]	MANET scheduling optimization	Significant enhancements in essential performance indicators, including network throughput, latency, and power consumption.
IoT and ADRIN2.0 [87]	Enhancing data forwarding post-disaster	Enhancing data transmission rate, energy consumption, and reducing latency.

an innovative method that leverages nature-inspired scheduling algorithms, primarily relying on observing a neutral system, to improve scheduling performance in dynamic and decentralized MANETs. For effective post-disaster message exchange, an adaptive and decentralized routing mechanism called ADRIN is introduced [87]. By sensing human movement, the proposed mechanism allows for effective data forwarding, which strikes a balance between data transmission rate and energy consumption while reducing latency.

However, exploiting MANET topology characteristics enables appropriate emergency communication. Incorporating various technologies, techniques, and protocols boosts the relief operation. Table 4 presents a succinct summary of various related technologies and techniques implemented in MANET.

In a crisis, the incorporation of VANET with cutting-edge technologies to bolster relief efforts is expected to preserve lives. Nevertheless, the emergency strategy hinges on VANET's ability to employ video streaming for precise vehicle routes, particularly for emergency vehicles. The authors in [88] proposed a model that employs visual sensors, cameras, and OBU to securely broadcast emergency video to connected nodes via 5G wireless technology. A further study [89] concentrated on the development of a unique technique known as received signal strength (RSS), an alternate scheme for Global Positioning Systems (GPS) to establish communication among vehicles through roadside RSUs in the metropolitan region. However, the proposed technique outperforms well-established least squares (LS) and weighted least squares methods (WLS) techniques in terms of dynamic vehicle ranging, ranging error, and throughput.

A novel online dynamic population evacuation (DPE) model addresses the challenge of the shelter allocation problem (SAP) in a real evacuation scenario. The model tackles the SAP by applying destination selection and dynamic traffic assignment techniques. The model under consideration investigated two distinct VANET topology configurations, including vehicular edge computing (VEC) and vehicular cloud computing (VCC). Compared to other models, it exhibited superior performance in terms of network clearance time, end-to-end delay, and packet delivery ratio. The model was discussed in detail in [90]. The authors conducted a study in [91] with the specific goal of enhancing the accuracy of V2V communication by leveraging the 5G New Radio (NR) system. The authors also used model-lane changes (IDM-LC) and intelligent driver model avoidance (IDM-A) to investigate how vehicles move in a variety of road situations. The findings indicated that the suggested IDM models showed feasibility and effectiveness.

Data dissemination in VANETs poses a critical challenge in guaranteeing a superior quality of service and preventing unsafe situations. These issues arise due to congestion or a

TABLE 5. Various related technologies and techniques implemented in VANET.

Technologies / Techniques	Primary objective	Merits
Sensors, cameras, OBU, and 5G wireless technology [88]	Determine emergency vehicle pathways	Securely broadcast emergency video.
The RSS-based localization algorithm [89]	Establishing communication through roadside RSUs	An alternate scheme for GPS. Enhancing dynamic vehicle ranging, ranging error, and throughput.
VEC and VCC [90]	Addressing dynamic population evacuation (DPE) problems	Enabling dynamic selection of a destination and traffic assignment.
5G NR system, IDM-LC, and IDM-A [91]	Enhancing the performance of V2V communication in VANET	Examining several road scenarios
HTEMD model [92]	Improving emergency message dissemination	Demonstrating a considerably higher rate of correct decision and resistance to attack.
CluRMA [93]	Enhancing message aggregation	Identifying safety and non-safety messages
ASPBT [94]	Establishing a reliable and accurate emergency communication broadcast system	Avoiding a broadcast storm and message delivery failure.
PSO and TMR [95]	Addressing broadcast storms in VANET	Enhancing throughput and packet loss ratio, as well as reduced end-to-end delay

broadcast storm and addressing them in VANET architectures necessitates implementing proper techniques. The main goal of the research in [92] is to implement a hybrid-trust-based emergency message dissemination (HTEMD) model that will make the VANET topology more flexible and reliable. The proposed model exhibits significant performance compared to the baseline model, demonstrating a considerably higher rate of correct decision and resistance to attack.

The authors in [93] implement two types of message aggregation by leveraging a cluster-based RSU-enabled message aggregation protocol (CluRMA). The message aggregation type includes local aggregation at the cluster heads and global aggregation at the RSU. Additionally, the authors performed the aggregation process using adaptive Huffman compression and arithmetic coding techniques to identify safety and non-safety messages, respectively. In [94], the authors developed the Adaptive Scheduled Partitioning and Broadcasting Technique (ASPBT) to avoid a broadcast storm and message delivery failure. The results show that VANET has an accurate and reliable emergency communication broadcast mechanism based on the proposed technique. In, [95], researchers integrated a meta-heuristic search approach with particle swarm optimization (PSO) and time delay-based multipath routing (TMR) to tackle broadcast storms. The proposed approach offers significant improvements in terms of throughput and packet loss ratio, as well as reduced end-to-end delay.

However, vehicles can efficiently share emergency messages by exploiting VANET topology characteristics and appropriate protocols. Fusing various technologies, techniques, and protocols raises emergency response operations. Table 5 briefly describes various relevant technologies and techniques employed in VANET.

A natural disaster caused significant damage to the pre-established communication infrastructure, which instantly resulted in a lack of communication between victims and rescue teams. Consequently, innovative technology like drones offers effective methods to maintain communication amidst a crisis. Nonetheless, implementing a well-designed FANET topology may mitigate the residual impacts of the disaster, considering certain obstacles such as unexpected failures, communication capacity, and power availability. Therefore, resolving previous obstacles directly influences the disaster management methodology and protects human life. In [96], the researchers conducted a study to improve reliable, self-aware, multi-hop UAV operation management to boost communication under infrastructure damage and overload situations. Furthermore, the researchers suggested a UAV operation management framework to allow optimal mission-orientated coverage. The enhancement analysis demonstrates increased flight periods, compatibility with overlay networks, and reliable service supply.

The research in [97] focuses particularly on the implementation of the Deep Deterministic Policy Gradient (DDPG) method to enable spatiotemporal dynamic deployment of air-ground ANET (AGANET) nodes. The suggested ad hoc network configuration comprises unmanned aerial vehicles (UAV) and ground-based nodes. Compared with traditional

TABLE 6. Various related technologies and techniques implemented in FANET.

Technologies / Techniques	Primary objective	Merits
A UAV operation management framework [96]	Boosting communication in the face of infrastructure damage and overload situation	Improving reliable, self-aware, multi-hop UAV operation management. Allowing optimal mission-orientated coverage.
DDPG, (UAV), and ground-based nodes [97]	Enabling spatiotemporal dynamic deployment of air-ground ANET (AGANET) nodes	Enhancing communication reliability, efficiency, and reduced costs in emergency events.
CUCO and PSO algorithms [98]	Handling cluster size issues	Improving in both throughput and packet dropping rate.
SDR-enabled FANET architecture, 3G/4G/5G/Wi-Fi systems, Software Defined Radios (SDRs), and HCM [99]	Enabling features like self-forming, self-organising, and self-healing capabilities	Enriching the communication among UAVs, mobile-to-UAV, and UAV-to-mobile.
EPSO model [100]	Improving efficiency, stability, and reliability in emergency monitoring	Facilitating prompt and reliable services in times of calamity.
A multi-UAV, B5G [101]	Reducing the energy usage and boosting network performance	Enhancing throughput, reliability, and energy efficiency, and coverage

methods, the proposed approach affords superior communication reliability, enhanced efficiency, and reduced costs in emergency events. In [98], the authors introduced two separate meta-heuristic optimisation algorithms, the Cuckoo Search Algorithm (CUCO) and the Particle Swarm Algorithm (PSO), to provide an unusual strategy for handling cluster size issues. The suggested method analysis reveals the improvement in both throughput and packet dropping rate (PDR).

In [99], the authors proposed an SDR-enabled FANET architecture in disaster-stressed regions. To enable features like self-forming, self-organising, and self-healing capabilities, the proposed architecture utilises wireless technology equipment, comprising 3G/4G/5G/Wi-Fi systems and Software Defined Radios (SDRs). Additionally, the authors implemented a hybrid connectivity module (HCM) to enrich the communication among UAVs, mobile-to-UAV, and UAV-to-mobile. In [100], researchers investigate the evolution of the evolutionary particle swarm optimization (EPSO)-based emergency monitoring geospatial edge service chain within the emergency communications network. The experiment findings indicate notable improvements in efficiency, stability, and reliability in emergency monitoring, while the investigated method facilitates prompt and reliable services in times of calamity.

In high-risk scenarios, the integration of advanced technologies, such as unmanned aerial vehicles (UAVs) and beyond fifth-generation (B5G) networks, enhances the operational efficiency of search and rescue (SAR) teams. A study [101] developed a multi-UAV and SAR device-to-device communication model aimed primarily at enhancing disaster response efficiency in scenarios involving floods, earthquakes, and forest fires. Additionally, the proposed model incorporates a clustering approach to reduce the energy usage and boost network performance. The evaluation of the suggested model demonstrated enhanced network throughput, reliability, and energy efficiency, along with broad coverage of the disaster zone.

Nevertheless, adopting FANET properties may be a highly effective approach to protecting people during disasters. This is particularly true when infrastructure damage is permanent and irreversible. Furthermore, the integration of technologies, methodologies, and protocols in FANET aims to enhance the effectiveness of response operations. Table 6 provides a concise overview of pertinent technologies and techniques used in FANET.

Numerous innovative approaches have been introduced and examined using diverse technologies, such as maritime wireless communication, radio communication for signal transmission, Delft 3D, and Delft Dashboard. While a conventional tsunami early warning system that uses buoys equipped with sensors for emergency communication has some limitations, the incorporation of existing technology seeks to enhance communication between ships in the open ocean and ship-to-coastal radio stations. Researchers in [102], combined the preceding technologies with VHF radio communication technology to improve communication in the worst-case scenario. In [103], researchers deployed VHF and HF radio communication systems alongside the preceding technologies to boost the duration of signal propagation.

TABLE 7. Various related technologies and techniques implemented in SANET.

Technologies / Techniques	Primary objective	Merits
VHF radio communication [102]	Improve communication in the worst-case scenario	Improving reliable, self-aware, multi-hop UAV operation management. Allowing optimal mission-orientated coverage.
VHF and HF radio communication [103]	Boosting the duration of signal propagation	Enhancing communication reliability, efficiency, and reduced costs in emergency events.
UAV and a genetic algorithm [104]	Enhancing sea-based maritime communication	Enhancing sea-based maritime communication, including islands, large vessels, drones, and marine aircraft communications.
HAPS, 6G networks, renewable energy sources (RES), and battery energy storage systems (BESSs) [105]	Enabling the effectiveness of natural disaster response	The network's resilience.

Additionally, the cost of massive data transmission in satellite communication technology presents a significant obstacle to the SANET topology. The authors in [104] proposed a multi-objective optimization model that leverages the features of unmanned aerial vehicles. The goal of UAV integration with a genetic algorithm is to enhance sea-based maritime communication, including islands, large vessels, drones, and marine aircraft communications. Hence, the findings show the practicality and efficiency of the model.

In [105], the authors designed the SANET architecture, a space-air-ground-sea topology specifically designed to reduce the impact of earthquakes. Additionally, the suggested design incorporates several technologies, including high-altitude platform stations (HAPS), 6G networks, renewable energy sources (RES), and battery energy storage systems (BESSs), to boost the effectiveness of natural disaster response. The evaluation of the suggested topology reveals the network's resilience.

Nevertheless, leveraging SANET topology for emergency communication seems potentially a viable area for future study and development. Even though various novel technologies aid in deploying reliable and higher-performance networks, routing protocols continue to pose significant challenges in SANET. Table 7 provides a summary of the relevant technologies and methods used in SANET.

In a WSN communication network, which comprises one or more sensors, the integration of various technologies leads to enhanced communication during an emergency. Authors in [106] introduced Fi-Model, a wireless sensor network that leverages the IoT to enhance the monitoring, detecting, and emitting alert processes in fire hazard scenarios. The proposed model integrates the ESP8285 controller, a Cloud-Based API, sensor, and machine learning approaches to establish an effective access control system that facilitates fire accident management. The outcome indicated the outstanding efficacy of the model in protecting lives and assets.

Another study [107] particularly concentrates on offline and online activity assignments for multiple UAVs in wireless sensor networks. For the offline activity assignments phase, authors enhance the immune multi-agent algorithm, resulting in higher reliability and convergence efficiency. They enhance adaptive discrete cuckoo algorithms, which directly impact real-time online activity assignments. The study by [108] introduced the intelligent healthcare system, leveraging nanosensor features to improve emergency communication between victims and clinical data servers. While the primary goal of deployable nanosensors is to gather health indicators in real-time, the responsibility of securely safeguarding victim data falls to genetic-based encryption techniques. The proposed intelligent system exhibits superior energy efficiency, reduced time consumption, and enhanced security.

The authors in [109] offered an empirical model that fuses the WSN, based on IoT, with artificial intelligence to develop an early warning system to deal with forest fires. The researcher developed the FFDNet technique for forest fire detection by incorporating image sensors and deep learning (DL) approaches. However, following the collection of images by sensors assembled in the base station, the FFDNet technique employed the kernel extreme learning machine (KELM) model for the classification. However, the FFDNet approach utilises the kernel extreme learning machine (KELM) model to classify data acquired by sensors at the base station. The assessments indicate improvements in the FFDNet technique compared to other deep learning models.

TABLE 8. Various related technologies and techniques implemented in WSN.

Technologies / Techniques	Primary objective	Merits
ESP8285 controller, a Cloud-Based API, sensor, and machine learning approaches [106]	Enhancing the monitoring, detecting, and emitting alert processes	Improving the performance of control systems to track and manage fire.
UAVs, multi-agent and adaptive discrete cuckoo algorithms [107]	Enhancing offline and online activity assignments	Boosting the acquisition of sensor data from extensive wireless sensor nodes following seismic events.
Nanosensors and genetic-based encryption [108]	Developing an intelligent healthcare system	Enhancing the collection of real-time health indicators.
KELM and image sensors [109]	Developing early-warning methods to deal with forest fires	Enhancing the noise removal process by the guided filtering (GF) technique.

However, the deployment of WSNs based on IoT topology for emergency communication offers a promising opportunity to improve pre- and post-disaster management procedures. Deploying WSNs in remote areas necessitates improving the network performance matrix by utilizing a variety of routing protocols and technologies. Table 8 presents a summary of the pertinent technologies and methods used in WSN.

Extraordinary natural occurrences pose distinct obstacles to communication. Harnessing the mobile network infrastructure for modern emergency rescue missions is considered a feasible method. The study referenced in [110] conducted a thorough evaluation of the intelligent crisis management system based on deep learning. The proposed smart system provides a means to safeguard human life during pandemics like the COVID-19 epidemic. The proposed system consists of several phases. In the initial phase, the Wireless Body Sensor Network (WSN) and hospitals provide individual medical records. The second phase uses the Kalman method to filter data, while the final phase applies the neural network-long short-term memory (CNN-LSTM) algorithm to issue emergency alerts. Moreover, the network architecture consists of a wireless access vehicular environment (WAVE) and the fifth generation of wireless cellular technology (5G) connected with IES-equipped devices, including ambulances and helicopters.

Researchers in [111] proposed a comprehensive approach for emergency communication known as Open6GRAN. The proposed architectural framework is an innovative method that seeks to create robust, durable, flexible, and open-source sixth-generation (6G) radio access networks with suitable communication capabilities to withstand natural disasters. The researcher integrated an artificial intelligence-based radio access network (RAN) controller to enable various services and swift recovery capabilities, while they deployed aerial and space segments with the terrestrial network to accommodate emergency communication. Nevertheless, the findings demonstrate significant improvements in performance during emergencies.

In [112], researchers proposed an emergency communication approach, namely RESPOND-A, designed to facilitate the operation of first responders. The proposed communication architecture consists of a cellular 5G network and Wi-Fi 6 relay devices mounted on UAVs, enabling direct flight to designated destinations and boosting communication. The suggested method successfully achieved high bit rates and fulfilled the first mission communication requirements by establishing a direct link between the 5G base station and users' equipment.

Ultimately, incorporating cellular networks with diverse network topologies represents substantial progress in emergency communication. The benefit stems from the permanent infrastructure, high mobile density, resource allocation, and other related factors. Table 9 presents a summary of the pertinent technologies and methods used in cellular networks.

One intriguing and promising approach in the domain of satellite communication involves the integration of several network topologies, such as FANET and software-defined satellite networks, to refine communication after a disaster. Authors in [113] pointed out the significance of combining the network of unmanned aerial vehicles (UAVs) and low Earth orbit (LEO) satellites. To mitigate issues related to received signal strength, data rate, and latency, the authors introduce a vertical HO scheme that employs Analytic Hierarchy Process-Entropy (AHP-Entropy) weighting and Technique for Order Preference by Similarity to the Ideal Solution (TOPSIS). While the AHP-Entropy method is responsible for classification and selecting the optimal network, mobile edge computing (MEC) on the LEO satellite is duty-bound for data compression to further reduce the LEO satellite feeder link delay. Nevertheless, experiment findings indicate that the introduced approach surpasses the benchmark method in terms of throughput and reduced disconnections.

TABLE 9. Various related technologies and techniques implemented in cellular networks.

Technologies / Techniques	Primary objective	Merits
UAVs, TOPSIS , and MEC [113]	Enhancing the stability and reliability of real-time post-disaster	Incorporating (UAVs) and (LEO) satellites topologies.
QZSS, and EWS [114]	Enhancing the dissemination of warning messages	Providing alternatives to ground-based networks in remote regions.
6G, IoT services, and UAVs [115]	Facilitating emergency communication across boundaries	Supporting the resource allocation in real-time and efficient control of network functions.

In [114], the author suggested harnessing the Quasi-Zenith Satellite System (QZSS) to develop the effectiveness of the early warning system (EWS). The traditional ground-based network currently restricts the dissemination of warning messages in regions affected by severe catastrophic events, whereas the suggested approach aims to increase this capacity and guarantee emergency communication. Furthermore, [115] examined the incorporation of non-terrestrial networks (NTNs) into 6G telecommunications to guarantee communication across boundaries. Researchers exploit the combination of IoT services, UAVs, LEOs, MEOs, and GEOs to mitigate communication challenges by reducing latency, improving energy efficiency, and facilitating several services.

The research in [116] dedicated to introducing edge-intelligence-driven collaborative SGIN architecture. The introduced architecture layers consist of the sensing layer, the forwarding layer, the control layer, the intelligence layer, and the application layer, which collaborate to detect diverse measurements, alleviate network congestion, and deliver ubiquitous intelligence services. Concretely, a novel integrated learning framework based on convolutional neural networks (CNN) is formulated to regulate the computation of offloading and network functions. Consequently, the proposed smart system exhibits a superior average transmission rate compared to random scheduling (RS) and greedy policy solutions.

In [117], the authors introduced the centrally deep reinforcement learning-aided multi-node federated learning (CDRFL) architecture to foster resource allocation and energy efficiency. The research thoroughly investigated the validity of the graph permutation property in the context of the Satellite-Ground Integrated Networks (SGIN) graph architectures. Furthermore, the assessment of the study involves the demonstration of the upper bound of quantization error through comprehensive mathematical derivation. The analysis revealed superior performance in terms of average latency and energy consumption.

The deployment of space-air-ground integrated networks (SAGINs) offers a resilient and adaptable topology to address the limitations of conventional networks, especially in ensuring the delivery of essential services during natural disasters. In [118], the authors proposed a novel SAGIN architecture leveraging software-defined networking (SDN) and network function virtualisation (NFV) technologies to enhance network efficiency. The proposed architecture aims to enhance service function chains (SFCs). Furthermore, the proposed adaptable network architecture incorporates rate-adaptive SFC management and wireless resource allocation, which enable superior network performance in severe conditions. A thorough evaluation demonstrated the proposed architecture's effectiveness in managing SFC and significantly enhanced resource allocation in highly dynamic network topologies.

Eventually, the integration of satellite networks with diverse network topologies provides an optimal solution for preserving global communication during emergencies. The benefit lies in the absence of infrastructure, resource allocation, and other related aspects. Table 10 offers a succinct overview of pertinent technologies and techniques used in satellite networks.

Sustaining emergency cyber services across vast geographic regions is exceedingly challenging during a disaster. Different network topologies allow for covering several locations to facilitate communication through base stations, UAVs, satellites, and cellular channels. The high-coverage topologies outlined in [19], [94], [119], [120], and [121] are appropriate for emergency communication over a broad geographical area.

A significant consequence that authorities encounter is the interruption of infrastructure, which can lead to a breakdown in communication between victims and rescue crews. Furthermore, the network expansion demands consistent performance and a greater density of participating nodes to enable emergency communication. To guarantee high network capacity while preserving reliability and scalability, however, a range of high-performance network topologies introduced in [86], [87], [92], [97], and [111] are suitable for deploying in hard-hit regions.

Additionally, incorporating different topologies and technologies poses challenges in the deployment of costly infrastructures. To ensure optimal network performance, specific factors such as scale, complexity, and network type should be considered, especially in emergencies. References [79], [84], and [87] investigate affordable cost topologies. Table 11 lists

TABLE 10. Various related technologies and techniques implemented in satellite networks.

Technologies / Techniques	Primary objective	Merits
UAVs, TOPSIS , and MEC [109]	Enhancing the stability and reliability of real-time post-disaster	Incorporating (UAVs) and (LEO) satellites topologies.
QZSS, and EWS [110]	Enhancing the dissemination of warning messages	Providing alternatives to ground-based networks in remote regions.
6G, IoT services, and UAVs [111]	Facilitating emergency communication across boundaries	Supporting the resource allocation in real-time and efficient control of network functions.
SGINs, CNN [116]	Reducing network congestion and for providing intelligence services	Incorporating CNN to achieve a superior average transmission rate
SGINs, CDRFL [117]	Enhancing the average latency and energy consumption	Incorporating Federated Learning (FL) to enhance communication between nodes.
SAGIN , CDRFL [118]	Improving resource allocation in highly dynamic network topologies	Integration rate-adaptive SFC management and wireless resource allocation.

the characteristics of all topologies proposed in this survey to assist network engineers and researchers in selecting a suitable topology that may meet their objectives.

2) EVALUATION OF EXISTING PROTOCOLS

Deploying the emergency network topologies assessed in the previous section presents substantial challenges. Although emergency communication encompasses many participating nodes that transmit critical data in a short period, choosing the appropriate network topology with the appropriate protocol should consider various factors such as node mobility, type of data, energy consumption, security, and more.

One of MANET's inevitable problems is route discovery, which requires tailored solutions. The dynamic connection and free mobility of nodes, enabled by the MANET topology, directly influence the performance of the network. However, routing protocols present an optimal solution to enhance performance by detecting the routes during catastrophe events. The research reported in [119] provided a comprehensive overview of the protocol type and traits, including the Ad hoc On-demand Distance Vector Routing (AODV) protocol, the Destination-Sequenced Distance Vector (DSDV) routing algorithm, and the Dynamic Source Routing Protocol (DSR).

In [122], the authors present a novel hybrid approach that integrates two on-demand and one table-driven routing protocol to improve the discovery route during disasters. Additionally, the assessment revealed a higher network density in terms of end-to-end delay, routing load and throughput, and data packet delivery density. For the enhancing Link State Routing Protocol (OLSR) protocol, the extended osprey-aided optimized link state routing protocol (EO_OLSRP) method incorporates a deep learning model was implemented to examine routing. The suggested model demonstrates superior performance in terms of a packet-delivery ratio (PDR), average end delay, and throughput [123].

In contrast, the study conducted by [124] applied an Optimal Fuzzy Clustering and Trust-based Routing (OFC-TR) approach to reduce the consumption of energy and latency while simultaneously improving network security and lifespan. In [125], the authors carried out a study to investigate the effectiveness of data transmission using the adaptive GPSR with dynamic thresholds (AGDT) protocol. Nevertheless, the innovative method merges the Greedy Perimeter Stateless Routing (GPSR) with a dynamic threshold-based approach. The findings indicate that AGDT surpasses competing protocols in terms of packet delivery ratio, end-to-end delay, and throughput.

In the context of VANET topology, routing protocols are crucial for the real-time dissemination of emergency messages (EMs) among vehicles. However, broadcasting various types of EMs, such as videos, voices, traffic congestion details, etc., demanded effective routing protocols while considering the dynamic network topology, high node mobility, volume of data, and network resource constraints. In the VANET environment, properly crafted protocols contribute to superior network performance by reducing latency, increasing throughput, and end-to-end delay.

Several further research domains exist to improve VANET routing protocols. In [126], the researchers introduced the Multi-Path Transmission Protocol for Video Streaming (MPTP-VS) to enhance the transmission of live emergency video by leveraging fog computing architecture. Consequently, the experimental evaluation of the proposed protocol demonstrates that it decreases latency, boosts throughput, and enhances video streaming performance. In [127], authors proposed a geographic routing technique that relies on trusted nodes to improve EM exchange. While trusted nodes concentrate on evaluating link quality and node quality, the research results show a substantial performance enhancement in message delivery rate, end-to-end delay, and network throughput.

In [128], the author introduces the supercluster-based urban multi-hop broadcast and best forwarder selection protocol (UMB-BFS). The proposed protocol employs modified density peak clustering (MDPC) for the clustering process, while improved firefly optimisation (IFO) duty is to select the optimal path. However, the findings indicate outstanding results in Edge-RSU for 5G VANET, including transmission delay and packet delivery ratio. The research in [129], suggested a unique clustering technique, known as the Cluster-Based Protocol for Prioritized Message Communication in VANET. By eliminating any delay during exchange EMs, the suggested protocol permits the temporary storage of non-urgent data. Consequently, it facilitates the swift exchange of EMs among vehicles.

A study [130] presents a hybrid RF/VLC system to show an effective adaptive routing protocol for emergency messages when there are a lot of people using VANET. While the quickly emerging Visible Light Communications (VLC) offers complementary short-range connectivity with high bandwidth and low interference, the presented protocol increases security and performance during an emergency.

Deploying FANET plays an essential role in facilitating rescue efforts during catastrophes. However, employing the proper protocol to enhance connectivity in infrastructure damaged areas offers numerous advantages, including the ability to expand, reliability, fault handling, energy efficiency, localization, increased coverage, improved connectivity, and reduced delay [120].

The authors in [131] proposed the Geographic Drone-Based Route Optimization (GDRO) approach, a uniquely designed communication link to achieve high connection stability and optimize FANET network performance. Additionally, the authors incorporated various methods, including Geographical Graph-Based Mapping (GGM) and Global Positioning System (GPS). While the GGM facilitates reaching the anchor node, the GPS consistently transmits its location information to assist in determining an accurate location. The proposed approach outperformed existing models in disaster area networks in terms of improved communication efficiency, higher throughput, decreased end-to-end delay, and packet loss performance.

FANET networks encounter many challenges, such as dynamic topology, high node mobility, and low density. These issues necessitate the use of adaptive relaying to ensure stable routing. Researchers examined an energy-aware routing scheme based on a virtual relay tunnel (EARVRT) in [132]. In addition, the proposed protocol demonstrates its superiority over standard routing protocols in terms of delay, network longevity, and packet delivery speed.

Researchers in [133] designed the Q-learning-aided resilient routing protocol with hindsight pre-calculation (QR2HPC) to establish emergency networks based on a swarm of unmanned aerial vehicles. The proposed protocol aimed at eliminating the intricate computation of deep reinforcement learning (DRL), which is not appropriate for emergency relief scenarios. Hence, the experiment yielded improved system performances in terms of packet delivery ratio, end-to-end delay, throughput, and lifetime of the network.

The implementation of SANET-based ground systems to facilitate air-to-sea (A2S) communication necessitates the use of a variety of emerging technologies, including 5G and 6G wireless technologies, cloud and edge computing, and the maritime Internet of Things (MIoT). In addition, to improve A2S communication, carefully engineered routing protocols are required to prevent delays and ensure reliable wireless transmissions. However, several protocols have been examined in SANET topology, including end-to-end transmission, cluster routing, and opportunistic routing protocols [134].

The work in [135] investigated the opportunistic routing protocols (ORP) within the framework of remote maritime emergency circumstances, discussing the probability of effectively transmitting emergency communication. However, the assessment of crucial indicators, such as the average latency and delivery probability, exposed the limitations of the ORP in distant maritime regions. In [136], researchers investigated the Ad hoc On-Demand Distance Vector Routing (AODV) protocol for warship communication. They focused on two important matrices: the packet delivery ratio (PDR) and the end-to-end delay. Nevertheless, the outcome demonstrated the outperformance of AODV with structured movement, while limitations arise with random movement.

WSN networks possess a notable advantage in monitoring and supervising critical infrastructures (CIs). Nevertheless, implementing effective routing protocols is crucial to guarantee reliability, security, and operational efficiency in several WSN applications. Additionally, the purpose of implementing the wireless protocols in WSNs is to enable internet connectivity for numerous IoT devices and transfer data [137].

The authors in [138] introduced the cluster energy hop-based dynamic route selection (CEH-DRS) approach, drawing inspiration from the strategies employed by bee colony optimisers. The proposed approach facilitates the routing of data packets obtained from multiple sensors, while the clustering process is responsible for various characteristics, such as throughput, delay, and packet delivery ratio. However, the studies demonstrated a boost in network performance by attaining better throughput.

In [139], the author suggested an Urban Adaptive Location-based Routing Protocol (UALRP) that serves within the context of dynamic wireless sensor networks. Furthermore, the proposed algorithm incorporates real-time data analytics methods and adaptive machine learning models to improve routing decisions in perpetual motion networks. The evaluation of the suggested protocol showed improvements in the network performance matrix, including latency, throughput, and energy efficiency. In contrast, the study by [140] proposed an adaptive priority scheduling model to improve critical packet delivery time and reduce average data delay. The protocol's algorithm design relies on multilevel priority packet classification, preemptive packet queuing, and an adaptive criticality-based packet next-hop selection mechanism. An assessment of performance reveals that the model effectively mitigates data loss and prioritizes emergency data transport.

Promoting high-quality service (QoS) and quality of experience (QoE) is critical for effective communication among user equipment (UEs) in cellular networks and beyond, particularly in the context of Fifth-Generation New Radio (5G-NR). Ensuring efficient execution of handover processes effectively allows users to communicate quickly, reliably, and with low latency. However, to eliminate challenges in cellular networks, the successful implementation of routing protocols requires careful consideration of critical performance metrics such as handover ratio, handover failure, and radio link failure [121].

Researchers in [141] introduced the collaborative energy-efficient routing protocol (CEERP) based on the multi-objective improved seagull algorithm (MOISA) to enhance and evaluate network performance during 5G and 6G transmission. The proposed algorithm leverages the reinforcement learning technique (R.L.) to employ collected data at the sink node and selection cluster head. The evaluation of CEERP revealed an enhancement in network lifespan and energy efficiency compared to existing protocols.

In [142], researchers developed an efficient mobile gateway selection and discovery-based routing protocol in a heterogeneous network. Providing internet gateways through Long Term Evolution (LTE) technology poses various challenges, especially when considering the integration of cellular communication and VANET architecture. The implemented routing protocol facilitated the identification and selection of internet getaways via an efficient multiple metrics-based relay selection method. Nevertheless, the results in terms of packet delivery ratio, packet delay, and overhead meet expectations. In contrast, the study by [143] boosted cellular vehicle-to-everything (C-V2X) communications by introducing a routing-aware (RA) protocol that is combined with cluster-based routing and a geo-based RA method. The proposed method offers an efficient strategy for managing communication resources. However, the outcome evidences an enhancement in spectrum efficiency.

As three-dimensional hierarchical satellite network, which consists of different air and ground layers, encounters various obstacles related to air computing. Air computing enables several services to cater to end users' needs, such as sending real-time video and communicating in an emergency. Therefore, to improve QoS and QoE, intelligent routing protocols and dynamic capacity enhancement are required [144].

The authors in [145] assessed multi-hop routing in a multi-tier hybrid satellite-terrestrial relay network (HSTRN) to measure the reliability of multi-hop routing in satellite networks. The investigated stationary optimal priority technique demonstrated an improvement in interruption probability and low complexity. In [146], researchers introduced a novel constrained multi-agent reinforcement learning approach (CMADR) to enable seamless communication services in isolated regions. The assessment of the proposed routing protocol shows a decrease in packet delay, energy usage, and packet loss rate.

The integration of space-air-ground networks poses numerous issues, including resource allocation and the establishment of multiple paths. The difficulty resides in ensuring reliability, fault tolerance, and stable communication in satellite networks. In [147], the authors proposed an approach to address the prior difficulties in Low Earth Orbit (LEO) satellite topologies, termed the Relative Position-based Efficient Routing (RP-ER). The proposed approach formulates the relative position model based on the laws of satellite motion. Afterwards, the central satellite provides address allocation instructions, permitting other satellites to allocate independent addresses and concurrently create primary and backup routes. The results of the experiment reveal that the RP-ER surpasses the traditional approaches, such as a Distributed Address Assignment Mechanism (DAAM) and AODV, regarding address allocation and packet transmission.

During catastrophic circumstances, the network continuously delivers various types of data, encompassing video, audio, location, and notifications. Although real-time data directly affects rescue operations, protocols are responsible for controlling data packet routes to shorten the duration of the delay. Additionally, sustaining emergency communication requires exploiting network resources to avoid network overhead. The well-designed protocol results in enhanced network performance metrics, including processing, bandwidth, and handshakes. However, [26], [133], [142] concentrated their analysis on enhancing routing protocols to reduce delay and overhead.

On one hand, transmitting vital data successfully over the network after a disaster represents an additional obstacle. This difficulty stems from the large volume of data transmitted simultaneously, which presents a negative influence on the throughput indicator. On the other hand, the loss of one or more data packets may result in a substantial loss of critical information. The need to improve packet delivery ratios, however, stems from the profound influence on network performance. Routing techniques for enhancing throughput and packet ratio are presented in [129], [132], and [140].

The mobility of nodes and data transmission results in a significant addition to energy consumption. While energy constraints, particularly in mobile ad hoc networks, lead to deprivation of connectivity, exploiting routing protocol characteristics may offer an optimal solution. Additionally, the secure routing of packets is a crucial aspect that demands further improvement. Routing procedures are used to ensure the security of vital data, including personal information and medical records. However, [20], [23], [36], [125] evaluates a few security-related enhancement methods.

Nevertheless, Table 12 provides an overview of the criteria for selecting protocols according to network topologies and limitations, such as delay, overhead, throughput, packet ratio, energy, and security. Lastly, this survey offers several tables that provide sufficient information to assist researchers and network engineers in fulfilling their needs.

VI. CHALLENGES AND FUTURE DIRECTION

Deploying a disaster area network constitutes an effective strategy for handling a range of disasters. The solutions discussed in the preceding sections exhibit the potential of DAN topologies, technologies, and protocols, together with the respective implementation outcomes and unresolved issues. Furthermore, the recently proposed solutions leveraged cutting-edge architecture and technology, including drones, IoT devices, integration of various topologies, and enhanced protocols to ensure emergency communication. Exploiting the features of prior technologies and architectures poses a unique range of technical, security, deployment, and social issues. Accordingly, further refinements and evaluation are essential to ensure emergency communication in life-threatening scenarios. This section outlines the primary challenges encountered in the realm of DAN and explores potential solutions for the disaster management systems.

A. CHALLENGES

1) SCALABILITY CHALLENGES

Scalability represents the bottleneck of the current disaster area network. Disaster management generally necessitates a scalable network architecture design to support numerous users and mobile devices, depending on the extent and breadth of the affected area. Moreover, ensuring network performance and reliability becomes increasingly essential as the network expands, presenting a continual challenge.

2) TECHNOLOGICAL INTEROPERABILITY

Experts seek to develop a unified framework for the integration of diverse topologies and technologies, including drones, IoT devices, and space-air-ground networks. The significant interoperability challenges arise from the employment of heterogeneous devices, routing protocols, communication standards, and security schemes. However, establishing standards is a crucial aspect of addressing these challenges, which demand evolution for a cohesive disaster area network.

3) SECURE MODEL DESIGN

Designing a secure model is a prevalent concern in emergency communication. Delivering emergency services entails the transmission and storage of sensitive information, including medical records and victims' identification. Consequently, offering a secure model to protect the information of victims and authorities from cyberattacks, such as eavesdropping attacks, impersonation attacks, denial-of-service attacks, and message falsification attacks, poses major challenges.

4) ENERGY CONSTRAINTS

In extreme disasters, such as an earthquake, infrastructure is vulnerable to damage, leading to a shortage of power sources. Portable devices employed in disaster communication rely on limited power batteries, which pose difficulties in recharging or replacement. Thus, deploying energy-intensive protocols and hardware presents a barrier to sustaining network operations after a calamity.

5) DYNAMIC NETWORK TOPOLOGIES

The swift movement of nodes inside the catastrophe zone illustrates the very dynamic characteristics of disaster environments, including the mobility of portable devices and rescue teams. Moreover, frequent node mobility hinders the development of effective routing protocols and the maintenance of stable emergency services. The deployment of space-air-ground topologies offers a feasible solution to mitigate node mobility challenges by encompassing extensive disaster areas. The prior topology offers a promising solution; nonetheless, it encounters a substantial challenge stemming from insufficient technology interoperability and difficulties in predicting disastrous expansion.

6) DEPLOYMENT AND REPAIR RESTRICTIONS

The swift deployment of a disaster area network is vital yet poses multiple challenges. The challenges include damaged infrastructure, resource limitations, and the employment of autonomous systems. Consequently, leveraging low-altitude and high-altitude network capabilities could enhance rapid deployment and network repair during a crisis.

7) COST EFFECTIVENESS

Developing and deploying a cutting-edge network for disaster areas presents a financial obstacle, particularly for resource-limited regions. The space-air-ground integrated topology poses a significant cost challenge, involving complex technological and infrastructure requirements. Meanwhile, a three-layer network guarantees the delivery of proper emergency services. Conversely, ad hoc network architecture offers a cost-effective solution in crisis scenarios. However, balancing advanced features with cost-effectiveness remains a significant challenge, constraining broad deployment and scalability in resource-limited areas.

8) SEVERE ENVIRONMENTAL CIRCUMSTANCES

Diverse disaster forms, including collapsed buildings, flood zones, wildfires, and harsh weather, directly affect the network infrastructure. Consequently, unpredictable terrain and environmental factors obstruct reliable emergency communication and the delivery of emergency services during such events. Hence, severe environmental circumstances present an additional challenge to the deployment of disaster networks.

9) SOCIAL AND MORAL CONSIDERATIONS

The earlier proposed solutions neglect to address moral and societal aspects, which are among the most critical concerns. Disaster response technologies must address issues related to equitable access across all socio-economic statuses. Additionally, cultural sensitivities and rules associated with the

TABLE 11. The characteristics of all topologies proposed in this survey.

	Topology	Cost	Reliability	Coverage	Scalability
[76]	WMN	●	●	📍	-
[77]	WMN	◔	●	📍	◔
[78]	WMN	-	◑	🗺	◔
[79]	WMN	◑	●	🗺	◕
[80]	WMN	◑	●	🗺 🌏	●
[81]	MANET	◑	●	📍 🗺	◕
[82]	MANET	◔	◷	📍 🗺	◔
[83]	MANET	◔	●	📍	-
[84]	MANET	◔	●	🗺	●
[85]	MANET	◔	●	📍	●
[86]	MANET	◷	●	🗺 🌏	◕
[87]	MANET	◔	●	📍 🗺	-
[88]	VANET	-	-	🗺	-
[89]	VANET	◔	◕	🗺 🌏	●
[90]	VANET	◷	◑	🗺	-
[91]	VANET	◔	●	📍 🗺	●
[92]	VANET	◔	●	🗺	●
[93]	VANET	◔	◑	🗺 🌏	◕
[94]	VANET	◔	◕	🗺	-
[95]	VANET	◔	●	🗺	-
[96]	FANET	●	●	🗺	●
[97]	FANET	◔	●	🗺	●
[98]	FANET	-	◑	🗺	-
[99]	FANET	●	◕	🗺 🌏	●
[100]	FANET	●	●	🗺 🌏	●
[102]	SANET	-	◑	🗺	◔
[103]	SANET	◕	●	🗺	◔

TABLE 11. *(Continued.)* **The characteristics of all topologies proposed in this survey.**

	Topology	Cost	Reliability	Coverage	Scalability
[104]	SANET	High	-	Regional	High
[105]	SANET	High	High	Regional, Global	High
[106]	WSN	Low	High	Local, Global	-
[107]	WSN	-	Moderate to High	Regional	Moderate
[108]	WSN	Low	High	Local, Global	High
[109]	WSN	-	-	Regional	-
[110]	Cellular	Low to Moderate	High	Regional	High
[111]	Cellular	High	High	Local, Global	High
[112]	Cellular	Low to Moderate	High	Regional, Global	Moderate to High
[113]	Satellite	High	High	Global	High
[114]	Satellite	High	High	Global	High
[115]	Satellite	High	High	Global	High

High Low Moderate Low to Moderate Moderate to High Local Regiona Global

dissemination of personal information must be addressed in the context of disaster area networks.

B. FUTURE DIRECTION

1) STANDARD PROTOCOLS TO FOSTER INTEROPERABILITY

Different technologies and topologies are inevitably assimilated into a disaster management system. Thus, establishing common communication protocols and standards facilitates the smooth integration of heterogeneous systems. In a catastrophic event, the adopted protocols and standards should be designed to accommodate the variances across distinct topologies, including spatial, aerial, and terrestrial networks.

2) AI-POWERED ADAPTIVE NETWORKING

Generally, leveraging artificial intelligence and machine learning enhances disaster network management. Accordingly, AI and ML could be employed in various contexts to meet constantly changing conditions in disaster scenarios. Potential improvements based on AI and ML encompass enhancing routing, predicting failures, boosting security, and dynamically allocating resources.

3) SECURE AND TRUSTWORTHY MODEL DEVELOPMENT

There is a lack of reliable mechanisms for emergency communication. The absence of a trustworthy model arises from various challenges, including node mobility, resource constraints, and the integration of heterogeneous systems. Hence, establishing a secure system is essential for guaranteeing the confidentiality and reliability of data exchanges in disaster scenarios.

4) RESILIENT NETWORK ARCHITECTURES

Natural catastrophes occur abruptly without any prior notification. Consequently, deploying a resilient network architecture is a vital means to offer effective and robust emergency services. Furthermore, proper deployment of a resilient network entails adapting to dynamic topologies and harsh environmental circumstances.

5) NETWORK SCALABILITY APPROACHES

Future research should focus on developing an affordable, scalable approach to ensure broad adoption and applicability in resource-limited regions. A prospective approach could exploit open-source software and hardware characteristics to enhance scalability in disaster areas. Additionally, collaborative shared infrastructure models such as satellite connections, several ad hoc network topologies, and cloud-based services could be integrated to guarantee emergency communication.

6) EMPLOYING EDGE COMPUTING TECHNIQUES

Enhancing real-time communication and reducing latency present a promising method for the swift provision of

TABLE 12. Recommended protocols according to network topologies and limitations.

Protocol	Topology	Delay	Overhead	Throughpu t	Energy	Packet ratio	Security
[123]	MANET	✓	✓	✓	✗	✓	✓
[124]	MANET	✓	✓	✓	✓	✓	✓
[125]	MANET	✓	✗	✓	✗	✓	✗
[128]	VANET	✓	✓	✓	✗	✓	✗
[129]	VANET	✓	✗	✗	✗	✓	✗
[130]	VANET	✓	✓	✓	✗	✓	✓
[131]	FANET	✓	✓	✓	✓	✓	✗
[132]	FANET	✓	✓	✗	✓	✓	✗
[133]	FANET	✓	✓	✓	✗	✓	✗
[135]	SANET	✓	✗	✗	✗	✓	✗
[138]	WSN	✓	✓	✓	✓	✓	✗
[139]	WSN	✓	✓	✓	✓	✓	✗
[140]	WSN	✓	✗	✗	✗	✓	✗
[141]	Cellular	✓	✓	✓	✓	✓	✗
[142]	Cellular	✓	✓	✗	✗	✓	✗
[143]	Cellular	✗	✗	✗	✗	✗	✗
[145]	Satellite	✓	✓	✗	✗	✗	✗
[146]	Satellite	✓	✗	✗	✓	✓	✗

emergency services through the integration of several topologies and technologies. Additionally, adopting edge computing to process real-time data allows expedited decision-making near disaster zones.

7) LOW-COST ALTERNATIVE

Low-cost alternatives are essential for the growth and deployment of disaster area networks. Reducing the deployment cost of network infrastructure, including technologies and equipment, is crucial for the extensive adoption of disaster area topologies.

8) RENEWABLE ENERGY TECHNOLOGIES

Enhancing energy-efficient routing protocols offers a viable method to extend battery lifetime. Additionally, renewable energy sources play an essential role in regions affected by disaster. Entire network nodes must be engineered to operate under low-power conditions, regardless of the transferred data during emergencies, including videos, audio, and location.

9) HUMAN-CENTERED DESIGN

Effective disaster response necessitates leveraging every available method. Generally, a potential disaster response framework could be designed to provide emergency services without requiring advanced technical skills. Therefore, integrating a user-friendly interface and conducting a training program empowers the effort to safeguard individuals' lives.

10) POLICY AND REGULATORY FRAMEWORKS

Establishing a comprehensive global framework to govern the moral and equitable deployment of disaster area networks could effectively address social and cultural challenges. Fostering international collaboration contributes to regulating innovative solutions that provide robust, efficient, and impactful responses, ultimately safeguarding lives and boosting global disaster response endeavours.

VII. CONCLUSION

This paper provides a comprehensive review of the evolution of topologies, emerging technologies, routing protocols, challenges, and research trends related to disaster area networks. This review commences with a concise overview of various DAN network architectures, subsequently presenting relevant literature and a systematic classification of diverse DAN topologies. Moreover, this study investigated the incorporation of various topologies and technologies, including 5G, 6G, UAVs, satellites, and AI innovations, to meet the requirements of DAN. This review exhibits diverse techniques for boosting the performance matrix, such as throughput, delay, overhead, security, and energy consumption. Afterwards, a comparative assessment of contemporary network topologies and routing protocols is conducted to identify various gaps and solutions. Ultimately, an analysis of current research trends and unresolved gaps in the field is highlighted with the identification of potential avenues for future research.

This survey aims to provide valuable resources for researchers, network engineers, and policymakers in the domain of DAN and emergency communication. In addition, it provides a thorough analysis of the state-of-the-art disaster area networks and delineates the key challenges and research trajectories necessary. The hope is that this survey will inspire further investigation in this current area and expedite DAN's advancement.

ABBREVIATIONS LIST

Term	**Description**.
DAN	Disaster Area Network.
5G	Fifth generation.
6G	Sixth generation.
AI	Artificial intelligence.
IoT	The Internet of Things.
MANET	Mobile ad-hoc networks.
VANET	Vehicular ad-hoc networks.
WSN	Wireless sensor networks.
UAVs	Unmanned aerial vehicles.
ANN	Artificial neural network.
GPS	The global positioning system.
EEWS	The earthquake early warning systems.
ML	Machine learning.
LoRa	Long Range.
RSSI	The received signal strength indicator.
P2P	Peer-to-peer.
WMN	Wireless Mesh Network.
ABC	Artificial Bee Colony.
QoS	Quality of service.
QoE	Quality of experience.
AODV	Ad hoc On-demand Distance Vector routing protocol.
DSDV	Destination-Sequenced Distance Vector routing protocol.
DSR	Dynamic Source Routing Protocol..
RSUs	Roadside units.
OBUs	Onboard units.
V2V	Vehicle-to-vehicle.
V2R	Vehicle-to-roadside.
V2I	Vehicle-to-infrastructure.
DSRCs	Dedicated short-range communications.
VEC	Vehicular edge computing.
VCC	Vehicular cloud computing.
EMs	Emergency messages.
FANET	Fly Ad Hoc Network.
SANET	Sea Ad Hoc Network.
UMVs	Unmanned marine vehicles.
UUVs	Unmanned underwater vehicles.
USVs	Unmanned surface vehicles.
A2S	Air-to-sea.
MIoT	Maritime Internet of Things..
DL	Deep learning.
IoMT	Internet of Medical Things.
UEs	User equipment.
GEO	Ggeosynchronous equatorial orbit.
MEO	Medium earth orbit.
LEO	Low earth orbit.
SDN	Software-defined networking.
LTE	Long Term Evolution.
SGIN	Satellite-Ground Integrated Networks.
BPNN	Backpropagation Neural Networks.
CNN	Convolutional Neural Networks.
DSDV	Destination-Sequenced Distance Vector.
NFV	Network function virtualization.
CDRFL	Centrally deep reinforcement learning-aided multi-node federated learning.

REFERENCES

[1] D. Cicek and B. Kantarci, "Use of mobile crowdsensing in disaster management: A systematic review, challenges, and open issues," *Sensors*, vol. 23, no. 3, p. 1699, Feb. 2023, doi: 10.3390/s23031699.

[2] M. Matracia, N. Saeed, M. A. Kishk, and M.-S. Alouini, "Post-disaster communications: Enabling technologies, architectures, and open challenges," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 1177–1205, 2022, doi: 10.1109/OJCOMS.2022.3192040.

[3] M. Ouaissa, M. Ouaissa, S. E. Himer, and Z. Boulouard, "AI and IoT integration for natural disaster management: A comprehensive review and future directions," in *AI and IoT for Proactive Disaster Management*. Hershey, PA, USA: IGI-Global, 2024, pp. 1–16, doi: 10.4018/979-8-3693-3896-4.ch001.

[4] M. Safizadeh, M. H. Marzbali, A. Abdullah, and N. Z. Maliki, "Proposed flood evacuation routes for heritage areas based on spatial configuration analysis: A case study of Penang, Malaysia," *J. Facilities Manage.*, vol. 22, no. 2, pp. 295–309, Mar. 2024, doi: 10.1108/jfm-11-2021-0137.

[5] S. Prasanna, M. R. Lenka, and A. R. Swain, "A survey on routing protocols for disaster management," *Social Netw. Comput. Sci.*, vol. 5, no. 2, Jan. 2024, Art. no. 216, doi: 10.1007/s42979-023-02509-2.

[6] M. A. Al-Absi, A. A. Al-Absi, M. Sain, and H. Lee, "Moving ad hoc networks—A comparative study," *Sustainability*, vol. 13, no. 11, p. 6187, May 2021, doi: 10.3390/su13116187.

[7] L. Khaloopour, Y. Su, F. Raskob, T. Meuser, R. Bless, L. Janzen, K. Abedi, M. Andjelkovic, H. Chaari, P. Chakraborty, M. Kreutzer, M. Hollick, T. Strufe, N. Franchi, and V. Jamali, "Resilience-by-design in 6G networks: Literature review and novel enabling concepts," 2024, *arXiv:2405.17480*.

[8] S. K. Abid, N. Sulaiman, S. W. Chan, U. Nazir, M. Abid, H. Han, A. Ariza-Montes, and A. Vega-Muñoz, "Toward an integrated disaster management approach: How artificial intelligence can boost disaster management," *Sustainability*, vol. 13, no. 22, p. 12560, Nov. 2021, doi: 10.3390/su132212560.

[9] K. Sharma, D. Anand, M. Sabharwal, P. K. Tiwari, O. Cheikhrouhou, and T. Frikha, "A disaster management framework using Internet of Things-based interconnected devices," *Math. Problems Eng.*, vol. 2021, pp. 1–21, May 2021, doi: 10.1155/2021/9916440.

[10] S. M. Khan, I. Shafi, W. H. Butt, I. D. L. T. Diez, M. A. L. Flores, J. C. Galán, and I. Ashraf, "A systematic review of disaster management systems: Approaches, challenges, and future directions," *Land*, vol. 12, no. 8, p. 1514, Jul. 2023, doi: 10.3390/land12081514.

[11] M. J. Anjum, T. Anees, F. Tariq, M. Shaheen, S. Amjad, F. Iftikhar, and F. Ahmad, "SpaCE-AIR-Ground integrated network for disaster management: Systematic literature review," *Appl. Comput. Intell. Soft Comput.*, vol. 2023, pp. 1–20, Feb. 2023, doi: 10.1155/2023/6037882.

[12] P. Chen, L. Luo, D. Guo, G. Tang, B. Zhao, Y. Li, and X. Luo, "Why and how Lasagna works: A new design of air-ground integrated infrastructure," *IEEE Netw.*, vol. 38, no. 2, pp. 132–140, Mar. 2024, doi: 10.1109/MNET.2024.3350025.

[13] Q. Wang, W. Li, Z. Yu, Q. Abbasi, M. Imran, S. Ansari, Y. Sambo, L. Wu, Q. Li, and T. Zhu, "An overview of emergency communication networks," *Remote Sens.*, vol. 15, no. 6, p. 1595, Mar. 2023, doi: 10.3390/rs15061595.

[14] R. Agrawal, N. Faujdar, C. A. T. Romero, O. Sharma, G. M. Abdulsahib, O. I. Khalaf, R. F. Mansoor, and O. A. Ghoneim, "Classification and comparison of ad hoc networks: A review," *Egyptian Informat. J.*, vol. 24, no. 1, pp. 1–25, Mar. 2023, doi: 10.1016/j.eij.2022.10.004.

[15] Z. Nurlan, T. Zhukabayeva, M. Othman, A. Adamova, and N. Zhakiyev, "Wireless sensor network as a mesh: Vision and challenges," *IEEE Access*, vol. 10, pp. 46–67, 2022, doi: 10.1109/ACCESS.2021.3137341.

[16] N. M. Jovanovic, "Analyzing wireless mesh network using spectral graph theory," *Artif. Intell. Appl.*, vol. 2, no. 3, pp. 195–201, Feb. 2023. [Online]. Available: https://scholar.google.com/scholar?as_q=Analyzing+Wireless+Mesh+Network+Using+Spectral+Graph+Theory&as_occt=title&hl=en&as_sdt=0%2C31

[17] A. Singh, S. Singh, and S. Prakash, "Critical comparative analysis and recommendation in MAC protocols for wireless mesh networks using multi-objective optimization and statistical testing," *Wireless Pers. Commun.*, vol. 129, no. 4, pp. 2319–2344, Apr. 2023, doi: 10.1007/s11277-023-10228-3.

[18] S. Al Ajrawi and B. Tran, "Mobile wireless ad-hoc network routing protocols comparison for real-time military application," *Spatial Inf. Res.*, vol. 32, no. 1, pp. 119–129, Feb. 2024, doi: 10.1007/s41324-023-00535-z.

[19] S. Hemalatha, M. Rajasekaran, L. K. Sagar, C. R. Komala, G. N. S. Vijayakumar, A. Nageswaran, M. Syamala, and J. Deepa, "A review of power management approaches for mobile ad hoc networks," *J. Européen des Systèmes Automatisés*, vol. 57, no. 1, pp. 137–145, Feb. 2024, doi: 10.18280/jesa.570114.

[20] H. Zhao, F. Ji, Y. Wang, K. Yao, and F. Chen, "Space–air–ground–sea integrated network with federated learning," *Remote Sens.*, vol. 16, no. 9, p. 1640, May 2024, doi: 10.3390/rs16091640.

[21] M. Okpok and B. Kihei, "Challenges and opportunities for multimedia transmission in vehicular ad hoc networks: A comprehensive review," *Electronics*, vol. 12, no. 20, p. 4310, Oct. 2023, doi: 10.3390/electronics12204310.

[22] A. W. K. Al-Nasir and F. S. Mubarek, "Vehicular Ad-hoc networks (VANETs): A survey on connectivity," *AIP Conf. Proc.*, vol. 2793, no. 1, pp. 1–12, Aug. 2023, doi: 10.1063/5.0163036.

[23] F. Pasandideh, J. P. J. da Costa, R. Kunst, N. Islam, W. Hardjawana, and E. P. de Freitas, "A review of flying ad hoc networks: Key characteristics, applications, and wireless technologies," *Remote Sens.*, vol. 14, no. 18, p. 4459, Sep. 2022, doi: 10.3390/rs14184459.

[24] F. Pasandideh, J. P. J. D. Costa, R. Kunst, W. Hardjawana, and E. P. de Freitas, "A systematic literature review of flying ad hoc networks: State-of-the-art, challenges, and perspectives," *J. Field Robot.*, vol. 40, no. 4, pp. 955–979, Jun. 2023, doi: 10.1002/rob.22157.

[25] W. D. Paredes, H. Kaushal, I. Vakilinia, and Z. Prodanoff, "LoRa technology in flying ad hoc networks: A survey of challenges and open issues," *Sensors*, vol. 23, no. 5, p. 2403, Feb. 2023, doi: 10.3390/s23052403.

[26] I. Bae and J. Hong, "Survey on the developments of unmanned marine vehicles: Intelligence and cooperation," *Sensors*, vol. 23, no. 10, p. 4643, May 2023, doi: 10.3390/s23104643.

[27] J. Li, G. Zhang, C. Jiang, and W. Zhang, "A survey of maritime unmanned search system: Theory, applications and future directions," *Ocean Eng.*, vol. 285, Oct. 2023, Art. no. 115359, doi: 10.1016/j.oceaneng.2023.115359.

[28] S. Rajendran and N. Chenniappan, "A comprehensive survey on several fire management approaches in wireless sensor networks," *Bull. Electr. Eng. Informat.*, vol. 13, no. 2, pp. 947–954, Apr. 2024, doi: 10.11591/eei.v13i2.5833.

[29] K. Kaur, A. Kaur, K. Sarita, K. Saini, S. Kour, and S. Kaur, "Emerging applications and ongoing challenges in WSN assisted IoT networks," *World J. Adv. Eng. Technol. Sci.*, vol. 11, no. 2, pp. 501–507, Apr. 2024, doi: 10.30574/wjaets.2024.11.2.0143.

[30] G. H. Adday, S. K. Subramaniam, Z. A. Zukarnain, and N. Samian, "Investigating and analyzing simulation tools of wireless sensor networks: A comprehensive survey," *IEEE Access*, vol. 12, pp. 22938–22977, 2024, doi: 10.1109/ACCESS.2024.3362889.

[31] S. Islam, A. Z. Abdulsalam, B. A. Kumar, M. K. Hasan, R. Kolandaisamy, and N. Safie, "Mobile networks toward 5G/6G: Network architecture, opportunities and challenges in smart city," *IEEE Open J. Commun. Soc.*, vol. 6, pp. 3082–3093, 2025, doi: 10.1109/OJCOMS.2024.3419791.

[32] C.-F. Lin and S.-H. Chang, "Advanced mobile communication techniques in the fight against the COVID-19 pandemic era and beyond: An overview of 5G/B5G/6G," *Sensors*, vol. 23, no. 18, p. 7817, Sep. 2023, doi: 10.3390/s23187817.

[33] M. Banafaa, I. Shayea, J. Din, M. Hadri Azmi, A. Alashbi, Y. Ibrahim Daradkeh, and A. Alhammadi, "6G mobile communication technology: Requirements, targets, applications, challenges, advantages, and opportunities," *Alexandria Eng. J.*, vol. 64, pp. 245–274, Feb. 2023, doi: 10.1016/j.aej.2022.08.017.

[34] W. Jiang, "Software defined satellite networks: A survey," *Digit. Commun. Netw.*, vol. 9, no. 6, pp. 1243–1264, Dec. 2023, doi: 10.1016/j.dcan.2023.01.016.

[35] C. Raja, M. Ramachandran, K. Ramu, and C. Sivaji, "Radio resource management satellite communication networks MCDM method," *Comput. Sci., Eng. Technol.*, vol. 1, no. 2, pp. 23–33, Jan. 2024, doi: 10.46632/cset/1/2/4.

[36] M. K. Banafaa, Ö. Pepeoğlu, I. Shayea, A. Alhammadi, Z. A. Shamsan, M. A. Razaz, M. Alsagabi, and S. Al-Sowayan, "A comprehensive survey on 5G-and-beyond networks with UAVs: Applications, emerging technologies, regulatory aspects, research trends and challenges," *IEEE Access*, vol. 12, pp. 7786–7826, 2024, doi: 10.1109/ACCESS.2023.3349208.

[37] S. Guha, R. K. Jana, and M. K. Sanyal, "Artificial neural network approaches for disaster management: A literature review," *Int. J. Disaster Risk Reduction*, vol. 81, Oct. 2022, Art. no. 103276, doi: 10.1016/j.ijdrr.2022.103276.

[38] M. Abdalzaher, M. Krichen, D. Yiltas-Kaplan, I. B. Dhaou, and W. Adoni, "Early detection of earthquakes using IoT and cloud infrastructure: A survey," *Sustainability*, vol. 15, no. 15, p. 11713, Jul. 2023, doi: 10.3390/su151511713.

[39] Z. T. AlAli and S. A. Alabady, "A survey of disaster management and SAR operations using sensors and supporting techniques," *Int. J. Disaster Risk Reduction*, vol. 82, Nov. 2022, Art. no. 103295, doi: 10.1016/j.ijdrr.2022.103295.

[40] R. Chataut, M. Nankya, and R. Akl, "6G networks and the AI revolution—Exploring technologies, applications, and emerging challenges," *Sensors*, vol. 24, no. 6, p. 1888, Mar. 2024, doi: 10.3390/s24061888.

[41] Z. Lokmic-Tomkins, D. Bhandari, C. Bain, A. Borda, T. C. Kariotis, and D. Reser, "Lessons learned from natural disasters around digital health technologies and delivering quality healthcare," *Int. J. Environ. Res. Public Health*, vol. 20, no. 5, p. 4542, Mar. 2023, doi: 10.3390/ijerph20054542.

[42] M. A. B. Siddiki Abir, M. Z. Chowdhury, and Y. M. Jang, "Software-defined UAV networks for 6G systems: Requirements, opportunities, emerging techniques, challenges, and research directions," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 2487–2547, 2023, doi: 10.1109/OJCOMS.2023.3323200.

[43] R. Damaševičius, N. Bacanin, and S. Misra, "From sensors to safety: Internet of Emergency Services (IoES) for emergency response and disaster management," *J. Sensor Actuator Netw.*, vol. 12, no. 3, p. 41, May 2023, doi: 10.3390/jsan12030041.

[44] C. Cheng, W. Chen, Y. Li, Y. Ji, S. Niu, Y. Hou, Q. Guo, and X. Chai, "Analysis of earthquake emergency command system according to cloud computing methods," *IEEE Access*, vol. 9, pp. 146970–146983, 2021, doi: 10.1109/ACCESS.2020.3019833.

[45] H. Fesenko, O. Illiashenko, V. Kharchenko, I. Kliushnikov, O. Morozova, A. Sachenko, and S. Skorobohatko, "Flying sensor and edge network-based advanced air mobility systems: Reliability analysis and applications for urban monitoring," *Drones*, vol. 7, no. 7, p. 409, Jun. 2023, doi: 10.3390/drones7070409.

[46] A. E. C. Redondi, C. Innamorati, S. Gallucci, S. Fiocchi, and F. Matera, "A survey on future millimeter-wave communication applications," *IEEE Access*, vol. 12, pp. 133165–133182, 2024, doi: 10.1109/access.2024.3438625.

[47] C. Indu and K. Vipin, "3D deployment of multi-UAV network in disaster zones to achieve coverage and connectivity," in *Proc. Int. Conf. Cognit. Robot. Intell. Syst. (ICC - ROBINS)*, Apr. 2024, pp. 715–722, doi: 10.1109/icc-robins60238.2024.10534020.

[48] M. R. Rezaee, N. A. W. A. Hamid, M. Hussin, and Z. A. Zukarnain, "Comprehensive review of drones collision avoidance schemes: Challenges and open issues," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 7, pp. 6397–6426, Jul. 2024, doi: 10.1109/TITS.2024.3375893.

[49] I. Chandran and K. Vipin, "Multi-UAV networks for disaster monitoring: Challenges and opportunities from a network perspective," *Drone Syst. Appl.*, vol. 12, pp. 1–28, Jan. 2024, doi: 10.1139/dsa-2023-0079.

[50] C. Gruffeille, A. Perrusquía, A. Tsourdos, and W. Guo, "Disaster area coverage optimisation using reinforcement learning," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, Jun. 2024, pp. 61–67.

[51] M. S. Adam, R. Nordin, N. F. Abdullah, A. Abu-Samah, O. A. Amodu, and M. H. Alsharif, "Optimizing disaster response through efficient path planning of mobile aerial base station with genetic algorithm," *Drones*, vol. 8, no. 6, p. 272, Jun. 2024, doi: 10.3390/drones8060272.

[52] H. Lin, M. A. Kishk, and M.-S. Alouini, "Virtual backhaul connectivity for enhanced coverage in fiber-less areas," *IEEE Wireless Commun.*, vol. 31, no. 4, pp. 324–330, Aug. 2024, doi: 10.1109/MWC.012.2300371.

[53] Y. M. Balakrishna and V. Shivashetty, "Device-to-device based path selection for post disaster communication using hybrid intelligence," *Int. J. Electr. Comput. Eng.*, vol. 14, no. 1, p. 796, Feb. 2024, doi: 10.11591/ijece.v14i1.pp796-810.

[54] F. Kagai, P. Branch, J. But, R. Allen, and M. Rice, "Rapidly deployable satellite-based emergency communications infrastructure," *IEEE Access*, vol. 12, pp. 139368–139410, 2024, doi: 10.1109/access.2024.3465512.

[55] A. Alhammadi, A. Abraham, A. Fakhreddine, Y. Tian, J. Du, and F. Bader, "Envisioning the future role of 3D wireless networks in preventing and managing disasters and emergency situations," 2024, *arXiv:2402.10600*.

[56] G. Sun, L. He, Z. Sun, Q. Wu, S. Liang, J. Li, D. Niyato, and V. C. M. Leung, "Joint task offloading and resource allocation in aerial-terrestrial UAV networks with edge and fog computing for post-disaster rescue," *IEEE Trans. Mobile Comput.*, vol. 23, no. 9, pp. 8582–8600, Sep. 2024.

[57] A. Samarakkody, D. Amaratunga, and R. Haigh, "Technological innovations for enhancing disaster resilience in smart cities: A comprehensive urban scholar's analysis," *Sustainability*, vol. 15, no. 15, p. 12036, Aug. 2023, doi: 10.3390/su151512036.

[58] S. H. Alsamhi, F. A. Almalki, H. Al-Dois, A. V. Shvetsov, M. S. Ansari, A. Hawbani, S. K. Gupta, and B. Lee, "Multi-drone edge intelligence and SAR smart wearable devices for emergency communication," *Wireless Commun. Mobile Comput.*, vol. 2021, no. 1, Jan. 2021, Art. no. 6710074, doi: 10.1155/2021/6710074.

[59] W. Feng, Y. Wang, Y. Chen, N. Ge, and C.-X. Wang, "Structured satellite-UAV-terrestrial networks for 6G Internet of Things," *IEEE Netw.*, vol. 38, no. 4, pp. 48–54, Jul. 2024, doi: 10.1109/MNET.2024.3380052.

[60] O. Ledesma, P. Lamo, and J. A. Fraire, "Trends in LPWAN technologies for LEO satellite constellations in the NewSpace context," *Electronics*, vol. 13, no. 3, p. 579, Jan. 2024, doi: 10.3390/electronics13030579.

[61] Y. Zhang, Y. Hong, M. Guizani, S. Wu, P. Zhang, and R. Liu, "A multi-layer information dissemination model and interference optimization strategy for communication networks in disaster areas," *IEEE Trans. Veh. Technol.*, vol. 73, no. 1, pp. 1239–1252, Jan. 2024, doi: 10.1109/TVT.2023.3304707.

[62] K. Demir, V. Tumen, S. Kosunalp, and T. Iliev, "A deep reinforcement learning algorithm for trajectory planning of swarm UAV fulfilling wildfire reconnaissance," *Electronics*, vol. 13, no. 13, p. 2568, Jun. 2024, doi: 10.3390/electronics13132568.

[63] M. A. Soomro, M. I. Channa, S. Z. Nizamani, M. A. Bhutto, and A. A. Ghoto, "Distance and energy aware DEAODV routing protocol in disastrous situation," *VAWKUM Trans. Comput. Sci.*, vol. 12, no. 2, pp. 137–148, Nov. 2024, doi: 10.21015/vtcs.v12i2.1977.

[64] U. Aliyu, H. Takruri, M. Hope, A. H. Gidado, and H. A. Adamu, "Disaster scenario optimised link state routing protocol and message prioritisation," *IET Netw.*, vol. 13, nos. 5–6, pp. 395–412, Sep. 2024, doi: 10.1049/ntw2.12125.

[65] J. P. J. Peixoto, J. C. N. Bittencourt, T. C. Jesus, D. G. Costa, P. Portugal, and F. Vasques, "Exploiting geospatial data of connectivity and urban infrastructure for efficient positioning of emergency detection units in smart cities," *Comput., Environ. Urban Syst.*, vol. 107, Jan. 2024, Art. no. 102054, doi: 10.1016/j.compenvurbsys.2023.102054.

[66] Q. Xiao, J. Zhao, S. Feng, G. Li, and A. Hu, "Securing NextG networks with physical-layer key generation: A survey," *Secur. Saf.*, vol. 3, Jan. 2024, Art. no. 2023021, doi: 10.1051/sands/2023021.

[67] H. Li, J. Sun, and X. Ke, "AI-driven optimization system for large-scale Kubernetes clusters: Enhancing cloud infrastructure availability, security, and disaster recovery," *J. Artif. Intell. Gen. Sci. (JAIGS)*, vol. 2, no. 1, pp. 281–306, Feb. 2024. [Online]. Available: https://ojs.boulibrary.com/index.php/JAIGS

[68] M. G. Spina, M. Tropea, and F. De Rango, "SURA-LB: Software-defined IDS with UAV resource aware load-balancing in FANET disaster scenarios," *Comput. Commun.*, vol. 223, pp. 101–114, Jul. 2024, doi: 10.1016/j.comcom.2024.05.015.

[69] R. Gao, S. Jing, M. Li, Y. Sun, G. Si, and Y. Zhang, "Research on health evaluation method of data center security protection system," *Proc. SPIE*, vol. 13175, p. 72, Jun. 2024, doi: 10.1117/12.3032059.

[70] K. P. Kumar, B. R. Prathap, M. M. Thiruthuvanathan, H. Murthy, and V. J. Pillai, "Secure approach to sharing digitized medical data in a cloud environment," *Data Sci. Manage.*, vol. 7, no. 2, pp. 108–118, Jun. 2024, doi: 10.1016/j.dsm.2023.12.001.

[71] V. Cheimaras, S. Papagiakoumos, N. Peladarinos, A. Trigkas, P. Papageorgas, D. D. Piromalis, and R. A. Munteanu, "Low-cost, open-source, experimental setup communication platform for emergencies, based on SD-WAN technology," *Telecom*, vol. 5, no. 2, pp. 347–368, May 2024, doi: 10.3390/telecom5020018.

[72] E. Cadet, O. S. Osundare, H. O. Ekpobimi, Z. Samira, and Y. W. Weldegeorgise, "AI-powered threat detection in surveillance systems: A real-time data processing framework," *Open Access Res. J. Eng. Technol.*, vol. 7, no. 2, pp. 31–45, Oct. 2024, doi: 10.53022/oarjet.2024.7.2.0057.

[73] M. Rihan, D. Wübben, A. Bhattacharya, M. Petrova, X. Yuan, A. Schmeink, and A. Fellan, "Unified 3D networks: Architecture, challenges, recent results, and future opportunities," *IEEE Open J. Veh. Technol.*, vol. 6, pp. 170–201, 2024, doi: 10.1109/OJCOMS.2024.011100.

[74] O. Deborah Segun-Falade, O. Soji Osundare, W. Edgar Kedi, P. Azuka Okeleke, T. Ignatius Ijomah, and O. Yetunde Abdul-Azeez, "Developing crossplatform software applications to enhance compatibility across devices and systems," *Comput. Sci. IT Res. J.*, vol. 5, no. 8, pp. 2040–2061, Aug. 2024, doi: 10.51594/csitrj.v5i8.1491.

[75] S. Hafeez, R. Cheng, L. Mohjazi, Y. Sun, and M. Ali Imran, "Blockchain-enhanced UAV networks for post-disaster communication: A decentralized flocking approach," 2024, *arXiv:2403.04796*.

[76] C.-W. Liang, Y.-L. Wu, C.-Y. Shi, S.-M. Lu, and H.-C. Lee, "Evaluation of a LoRA mesh wireless networking system supporting time-critical transmission and data lost recovery," in *Proc. 18th Int. Conf. Inf. Process. Sensor Netw.*, Apr. 2019, pp. 317–318, doi: 10.1145/3302506.3312607.

[77] B. Black, J. Rafferty, J. Santos, A. Ennis, P. Perry, and M. McKee, "Bluetooth 5.0 suitability assessment for emergency response within fire environments," *Electronics*, vol. 12, no. 22, p. 4599, Nov. 2023, doi: 10.3390/electronics12224599.

[78] R. Iqbal, A. Kashif, O. Alharbi, S. Zafar, A. Aljaedi, and Y. Qasaymeh, "Robustness in mesh networks using connected safe set and applications," *IEEE Access*, vol. 12, pp. 18021–18027, 2024, doi: 10.1109/ACCESS.2024.3357623.

[79] A. Chapnevis and E. Bulut, "UAV mesh network trajectory planning for age optimal data collection in infrastructureless areas," in *Proc. IEEE Int. Conf. Commun.*, Jun. 2024, pp. 1563–1568.

[80] M. Tham, Y. J. Wong, B.-H. Kwan, X. H. Ng, and Y. Owada, "Artificial intelligence of things (AIoT) for disaster monitoring using wireless mesh network," in *Proc. 6th Int. Conf. Softw. Eng. Inf. Manage. (ICSIM)*, Palmerston North, New Zealand, Jan. 2023, pp. 234–239, doi: 10.1145/3584871.3584905.

[81] D. Thaha, M. V. Lakshmi, S. Swathi, S. A. Sabatini, and T. Manikandan, "MANET based integrated sensor system for disaster detection and communication in hazardous environments," *Nanotechnology*, vol. 20, no. S7, pp. 82–92, 2024. [Online]. Available: http://www.nano-ntp.com

[82] J. Höchst, L. Baumgärtner, F. Kuntke, A. Penning, A. Sterz, M. Sommer, and B. Freisleben, "Mobile device-to-device communication for crisis scenarios using low-cost LoRA modems," in *Disaster Management and Information Technology* (Public Administration and Information Technology), vol. 40. Cham, Switzerland: Springer, Jan. 2023, pp. 235–268, doi: 10.1007/978-3-031-20939-0_12.

[83] N. Verma, M. Varshney, and N. Verma, "Role of MANET and artiicial bee colony (ABC) algorithm to make smart campus model, with smart sensors," *Res. Square*, vol. 2023, pp. 1–19, Jun. 2023, doi: 10.21203/rs.3.rs-3075962/v1.

[84] A. Abu Salem, "An effective management model for data caching in MANET environment," *Int. Arab J. Inf. Technol.*, vol. 20, no. 6, pp. 841–851, Nov. 2023, doi: 10.34028/iajit/20/6/1.

[85] I. Ambika, S. Bhatia, S. Basheer, and P. Dadheech, "Optimized resource allocation and queue management for traffic control in MANET," *Comput. Syst. Sci. Eng.*, vol. 45, no. 2, pp. 1323–1342, 2023, doi: 10.32604/csse.2023.030786.

[86] S. K. A. S. Syed Wajahat Abbas Rizvi, "Enhancing MANET performance: A novel approach through nature-inspired scheduling algorithms," *J. Electr. Syst.*, vol. 20, no. 3s, pp. 2030–2038, Mar. 2024.

[87] S. Halder, S. Roy, P. Ghosh, and N. Ghosh, "ADRIN2.0: Enabling post-disaster communication through adaptive mobility-informed routing," *IEEE Access*, vol. 12, pp. 102368–102380, 2024, doi: 10.1109/ACCESS.2024.3432866.

[88] M. Awais, Y. Saeed, A. Ali, S. Jabbar, A. Ahmad, Y. Alkhrijah, U. Raza, and Y. Saleem, "Deep learning based enhanced secure emergency video streaming approach by leveraging blockchain technology for vehicular AdHoc 5G networks," *J. Cloud Comput.*, vol. 13, no. 1, Aug. 2024, Art. no. 130, doi: 10.1186/s13677-024-00665-1.

[89] W. Ahmad, G. Husnain, S. Ahmed, F. Aadil, and S. Lim, "Received signal strength-based localization for vehicle distance estimation in vehicular ad hoc networks (VANETs)," *J. Sensors*, vol. 2023, no. 1, Jan. 2023, Art. no. 7826992, doi: 10.1155/2023/7826992.

[90] H. Idoudi, M. Ameli, C. Nguyen Van Phu, M. Zargayouna, and A. Rachedi, "Smart dynamic evacuation planning and online management using vehicular communication system," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 39, no. 10, pp. 1452–1468, May 2024, doi: 10.1111/mice.13148.

[91] R. Ali, R. Liu, A. Nayyar, I. Waris, L. Li, and M. A. Shah, "Intelligent driver model-based vehicular ad hoc network communication in real-time using 5G new radio wireless networks," *IEEE Access*, vol. 11, pp. 4956–4971, 2023, doi: 10.1109/ACCESS.2023.3234964.

[92] J. Qi, N. Zheng, M. Xu, P. Chen, and W. Li, "A hybrid-trust-based emergency message dissemination model for vehicular ad hoc networks," *J. Inf. Secur. Appl.*, vol. 81, Mar. 2024, Art. no. 103699, doi: 10.1016/j.jisa.2024.103699.

[93] B. Bhabani and J. Mahapatro, "CluRMA: A cluster-based RSU-enabled message aggregation scheme for vehicular ad hoc networks," *Veh. Commun.*, vol. 39, Feb. 2023, Art. no. 100564, doi: 10.1016/j.vehcom.2022.100564.

[94] M. R. Devi, I. J. S. Jeya, and S. Lokesh, "Adaptive scheduled partitioning technique for reliable emergency message broadcasting in VANET for intelligent transportation systems," *Automatika*, vol. 64, no. 2, pp. 341–354, Apr. 2023, doi: 10.1080/00051144.2022.2140392.

[95] D. Desai, H. El-Ocla, and S. Purohit, "Data dissemination in VANETs using particle swarm optimization," *Sensors*, vol. 23, no. 4, p. 2124, Feb. 2023, doi: 10.3390/s23042124.

[96] M. A. Abdel-Malek and M. Azab, "UAV-fleet management for extended NextG emergency support infrastructure with QoS and cost aware," *Internet Things*, vol. 25, Apr. 2024, Art. no. 101043, doi: 10.1016/j.iot.2023.101043.

[97] Y. Zeng, X. Tan, M. Sha, Z. Khadim Hussain, T. Lin, J. Tu, H. Wang, B. Liu, C. Li, F. Huang, and Z. Sha, "The study of DDPG based spatiotemporal dynamic deployment optimization of air-ground ad hoc network for disaster emergency response," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 128, Apr. 2024, Art. no. 103708, doi: 10.1016/j.jag.2024.103708.

[98] A. F. M. S. Shah, M. A. Karabulut, and K. Rabie, "Improvement of emergency communication systems using drones in 5G and beyond for safety applications," in *Proc. IEEE Future Netw. World Forum (FNWF)*, Nov. 2023, pp. 1–6.

[99] R. Rukaiya, S. A. Khan, M. U. Farooq, and I. Matloob, "Communication architecture and operations for SDR-enabled UAVs network in disaster-stressed areas," *Ad Hoc Netw.*, vol. 160, Jul. 2024, Art. no. 103506, doi: 10.1016/j.adhoc.2024.103506.

[100] S. He, X. Tan, Y. Zhong, M. Huang, Z. Mei, Y. Wan, and H. Wang, "Evolutionary PSO-based emergency monitoring geospatial edge service chain in the emergency communication network," *Int. J. Digit. Earth*, vol. 16, no. 1, pp. 2797–2817, Oct. 2023, doi: 10.1080/17538947.2023.2239765.

[101] A. Saif, K. Dimyati, K. A. Noordin, N. A. Mosali, and S. H. Alsamhi, "Skyward bound: Empowering disaster resilience with multi-UAV-assisted B5G networks for enhanced connectivity and energy efficiency," *Internet Things*, vol. 23, Oct. 2023, Art. no. 100885, doi: 10.1016/j.iot.2023.100885.

[102] M. B. Adityawan, A. K. Nurendyastuti, M. R. Purnama, M. S. Arifianto, M. Farid, A. A. Kuntoro, and Widyaningtias, "Development of a tsunami early warning system on the coast of Palu based on maritime wireless communication," *Prog. Disaster Sci.*, vol. 19, Oct. 2023, Art. no. 100290, doi: 10.1016/j.pdisas.2023.100290.

[103] A. K. Nurendyastuti, M. B. Adityawan, M. Rizki, M. S. A. Purnama, M. Farid, and A. A. Kuntoro, "A new approach for a tsunami early warning system based on maritime wireless communication, case study, pangandaran, Indonesia," *J. Appl. Sci. Eng.*, vol. 17, no. 4, pp. 2285–2294, 2023, doi: 10.6180/jase.202404_27(4).0007.

[104] X. Li and S. Jiang, "Deployment method for aircraft-based maritime emergency communication resource reserve bases," *J. Mar. Sci. Eng.*, vol. 12, no. 5, p. 844, May 2024, doi: 10.3390/jmse12050844.

[105] B. Karaman, I. Basturk, S. Taskin, F. Kara, E. Zeydan, and H. Yanikomeroglu, "Enhancing resiliency of integrated space-air ground-sea networks with renewable energies: A use case after the 2023 türkiye earthquake," *IEEE Commun. Mag.*, vol. 62, no. 12, pp. 104–111, Dec. 2024. [Online]. Available: https://blog.cloudflare.com/q1-2023-internet-disruption-summary/

[106] M. Ifeanyi Akazue, A. Efetobore Edje, M. D. Okpor, W. Adigwe, P. O. Ejeh, C. C. Odiakaose, A. A. Ojugo, E. B. Edim, R. E. Ako, and V. O. Geteloma, "FiMoDeAL: Pilot study on shortest path heuristics in wireless sensor network for fire detection and alert ensemble," *Bull. Electr. Eng. Informat.*, vol. 13, no. 5, pp. 3534–3543, Oct. 2024, doi: 10.11591/eei.v13i5.8084.

[107] L. Ye, Y. Yang, W. Meng, X. Wu, X. Li, and R. Zhu, "Offline and online task allocation algorithms for multiple UAVs in wireless sensor networks," *EURASIP J. Adv. Signal Process.*, vol. 2024, no. 1, Feb. 2024, Art. no. 21, doi: 10.1186/s13634-024-01116-4.

[108] T. Jabeen, I. Jabeen, H. Ashraf, N. Z. Jhanjhi, A. Yassine, and M. S. Hossain, "An intelligent healthcare system using IoT in wireless sensor network," *Sensors*, vol. 23, no. 11, p. 5055, May 2023, doi: 10.3390/s23115055.

[109] K. K. Paidipati, C. Kurangi, A. S. K. Reddy, G. Kadiravan, and N. H. Shah, "Wireless sensor network assisted automated forest fire detection using deep learning and computer vision model," *Multimedia Tools Appl.*, vol. 83, no. 9, pp. 26733–26750, Sep. 2023, doi: 10.1007/s11042-023-16647-5.

[110] H. Frikha, F. Kamoun-Abid, A. Meddeb-Makhoulf, and F. Zarai, "A smart emergency response system based on deep learning and Kalman filter: The case of COVID-19," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 34, no. 1, p. 630, Apr. 2024, doi: 10.11591/ijeecs.v34.i1.pp630-640.

[111] M. Yaser Yagan, S. Kayraklik, S. Kesir, G. Sumen, I. Hokelek, M. Basaran, G. C. Alexandropoulos, E. Basar, C. Cavdar, H. Arslan, and A. Gorcin, "Fast network recovery from large-scale disasters: A resilient and self-organizing RAN framework," 2024, *arXiv:2408.08609*.

[112] M. Batistatos, M.-A. Kourtis, G. Xilouris, D. Santorinaios, A. Oikonomakis, E.-Z. Bozis, and A. Kourtis, "Wi-Fi 6 aerial relay node in 5G network for emergency operations," *AEU - Int. J. Electron. Commun.*, vol. 170, Oct. 2023, Art. no. 154776, doi: 10.1016/j.aeue.2023.154776.

[113] K. Chen, L. Zhang, and J. Zhong, "Vertical handover strategy in satellite-aerial based emergency communication networks," in *Proc. 11th Int. Conf. Wireless Netw. Mobile*, vol. 46, Jul. 2024, pp. 1–6. [Online]. Available: https://eprints.whiterose.ac.uk/

[114] G. Potutan and K. Suzuki, "Addressing early warning challenges using satellites to improve emergency evacuation," *Emergency Manage. Sci. Technol.*, vol. 3, no. 1, pp. 1–10, 2023, doi: 10.48130/emst-2023-0004.

[115] K. Patil and B. Desai, "From remote outback to urban jungle: Achieving universal 6G connectivity through hybrid terrestrial-aerial-satellite networks," *Adv. Comput. Sci.*, vol. 6, no. 1, pp. 1–13, 2023.

[116] Y. Gong, H. Yao, and A. Nallanathan, "Intelligent sensing, communication, computation, and caching for satellite-ground integrated networks," *IEEE Netw.*, vol. 38, no. 4, pp. 9–16, Jul. 2024.
[117] Y. Gong, D. Yu, X. Cheng, C. Yuen, M. Bennis, and M. Debbah, "Computation offloading and quantization schemes for federated satellite-ground graph networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 10, pp. 14140–14154, Oct. 2024, doi: 10.1109/twc.2024.3409691.
[118] J. He, N. Cheng, Z. Yin, C. Zhou, H. Zhou, W. Quan, and X.-H. Lin, "Service-oriented network resource orchestration in space-air-ground integrated network," *IEEE Trans. Veh. Technol.*, vol. 73, no. 1, pp. 1162–1174, Jan. 2024, doi: 10.1109/TVT.2023.3301676.
[119] Y. Jahir, M. Atiquzzaman, H. Refai, A. Paranjothi, and P. G. LoPresti, "Routing protocols and architecture for disaster area network: A survey," *Ad Hoc Netw.*, vol. 82, pp. 1–14, Jan. 2019, doi: 10.1016/j.adhoc.2018.08.005.
[120] R. Dhall and S. Dhongdi, "Review of protocol stack development of flying ad-hoc networks for disaster monitoring applications," *Arch. Comput. Methods Eng.*, vol. 30, no. 1, pp. 37–68, Jan. 2023, doi: 10.1007/s11831-022-09791-y.
[121] A. Haghrah, M. P. Abdollahi, H. Azarhava, and J. M. Niya, "A survey on the handover management in 5G-NR cellular networks: Aspects, approaches and challenges," *EURASIP J. Wireless Commun. Netw.*, vol. 2023, no. 1, Jun. 2023, doi: 10.1186/s13638-023-02261-4.
[122] A. M. Soomro, A. B. Naeem, B. Senapati, K. Bashir, S. Pradhan, M. I. Ghafoor, and H. A. Sakr, "In MANET: An improved hybrid routing approach for disaster management," in *Proc. IEEE Int. Conf. Emerg. Trends Eng., Sci. Technol. (ICES&T)*, Jan. 2023, pp. 1–6, doi: 10.1109/ICEST56843.2023.10138831.
[123] H. A. Ahmed and H. A. A. Al-Asadi, "An optimized link state routing protocol with a blockchain framework for efficient video-packet transmission and security over mobile ad-hoc networks," *J. Sensor Actuator Netw.*, vol. 13, no. 2, p. 22, Mar. 2024, doi: 10.3390/jsan13020022.
[124] C. Edwin Singh, S. Sharon Priya, B. Muthu Kumar, K. Saravanan, A. Neelima, and B. Gireesha, "Trust aware fuzzy clustering based reliable routing in manet," *Meas., Sensors*, vol. 33, Jun. 2024, Art. no. 101142, doi: 10.1016/j.measen.2024.101142.
[125] A. Kumari, "AGDT: A mathematical model of hybrid algorithm for efficient routing in MANETs using dynamic thresholds," *Commun. Appl. Nonlinear Anal.*, vol. 31, no. 2s, pp. 72–83, May 2024.
[126] S. Benzerogue, S. Abdelatif, S. Merniz, S. Harous, and L. Khamer, "Multi-path transmission protocol for video streaming over vehicular fog computing environments," *IEEE Access*, vol. 12, pp. 87199–87216, 2024, doi: 10.1109/ACCESS.2024.3417288.
[127] B. Su and L. Tong, "Transmission protocol of emergency messages in VANET based on the trust level of nodes," *IEEE Access*, vol. 11, pp. 68243–68256, 2023, doi: 10.1109/ACCESS.2023.3292234.
[128] M. A. Albeyar, I. Smaoui, H. Mnif, and S. Alani, "Proposed supercluster-based UMBBFS routing protocol for emergency message dissemination in edge-RSU for 5G VANET," *Computers*, vol. 13, no. 8, p. 208, Aug. 2024, doi: 10.3390/computers13080208.
[129] S. Saha, V. V. Kumar, V. R. Niveditha, V. A. Kannan, K. Gunasekaran, and K. Venkatesan, "Cluster-based protocol for prioritized message communication in VANET," *IEEE Access*, vol. 11, pp. 67434–67442, 2023, doi: 10.1109/ACCESS.2023.3286936.
[130] N. Hassan, X. Fernando, and I. Woungang, "An emergency message routing protocol for improved congestion management in hybrid RF/VLC VANETs," *Telecom*, vol. 5, no. 1, pp. 21–47, Dec. 2023, doi: 10.3390/telecom5010002.
[131] V. Krishnakumar and R. Asokan, "Geographic drone-based route optimization approach for emergency area ad-hoc network," *Comput. Syst. Sci. Eng.*, vol. 45, no. 1, pp. 985–1000, 2023, doi: 10.32604/csse.2023.029189.
[132] M. Hosseinzadeh, S. Ali, A. H. Mohammed, J. Lansky, S. Mildeova, M. S. Yousefpoor, E. Yousefpoor, O. Hassan Ahmed, A. M. Rahmani, and A. Mehmood, "An energy-aware routing scheme based on a virtual relay tunnel in flying ad hoc networks," *Alexandria Eng. J.*, vol. 91, pp. 249–260, Mar. 2024, doi: 10.1016/j.aej.2024.02.006.
[133] J. Tang, Z. Zhou, W. Feng, and K. K. Wong, "A distributed and adaptive routing protocol for UAV-aided emergency networks," in *Proc. IEEE 98th Veh. Technol. Conf. (VTC-Fall)*, Oct. 2023, pp. 1–6.
[134] S. Liu, L. Zhu, F. Huang, A. Hassan, D. Wang, and Y. He, "A survey on Air-to-Sea integrated maritime Internet of Things: Enabling technologies, applications, and future challenges," *J. Mar. Sci. Eng.*, vol. 12, no. 1, p. 11, Dec. 2023, doi: 10.3390/jmse12010011.
[135] C. Liew and M. Radenkovic, "Exploring opportunistic routing for remote sea emergencies," 2024, *arXiv:2404.03013*.
[136] A. Jala Tirta Segara and A. D. Ramadhani, "Performance analysis of mobile ad-hoc networks based on TCP and UDP traffic on AODV protocol for warship communication," *J. Syst. Eng. Inf. Technol.*, vol. 2, no. 2, pp. 53–58, Sep. 2023, doi: 10.29207/joseit.v2i2.5343.
[137] S. Daousis, N. Peladarinos, V. Cheimaras, P. Papageorgas, D. D. Piromalis, and R. A. Munteanu, "Overview of protocols and standards for wireless sensor networks in critical infrastructures," *Future Internet*, vol. 16, no. 1, p. 33, Jan. 2024, doi: 10.3390/fi16010033.
[138] K. K. Almuzaini, S. Joshi, S. Ojo, M. Aggarwal, P. Suman, P. K. Pareek, and P. K. Shukla, "Optimization of the operational state's routing for mobile wireless sensor networks," *Wireless Netw.*, vol. 30, no. 6, pp. 5247–5261, Aug. 2024, doi: 10.1007/s11276-023-03246-3.
[139] O. I. Akinola, "Adaptive location-based routing protocols for dynamic wireless sensor networks in urban cyber-physical systems," *J. Eng. Res. Rep.*, vol. 26, no. 7, pp. 424–443, Jul. 2024, doi: 10.9734/jerr/2024/v26i71220.
[140] M. C. Pognon, A. Quintero, and S. Pierre, "Adaptive priority scheduling of Internet of Things data for disaster management in smart cities," *IEEE Access*, vol. 12, pp. 83285–83298, 2024, doi: 10.1109/ACCESS.2024.3407672.
[141] H. L. Gururaj, R. Natarajan, N. A. Almujally, F. Flammini, S. Krishna, and S. K. Gupta, "Collaborative energy-efficient routing protocol for sustainable communication in 5G/6G wireless sensor networks," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 2050–2061, 2023, doi: 10.1109/OJCOMS.2023.3312155.
[142] D. Abada, R. Adrdor, O. Boutkhoum, and A. Bohouch, "An efficient mobile gateway selection and discovery based-routing protocol in heterogeneous LTE-VANET networks," *Int. J. Comput. Netw. Commun.*, vol. 15, no. 2, pp. 57–79, Mar. 2023, doi: 10.5121/ijcnc.2023.15204.
[143] J. S. Alrubaye and B. S. Ghahfarokhi, "Geo-based resource allocation for joint clustered V2I and V2V communications in cellular networks," *IEEE Access*, vol. 11, pp. 82601–82612, 2023, doi: 10.1109/ACCESS.2023.3300294.
[144] B. Yamansavascilar, A. Ozgovde, and C. Ersoy, "Air computing: A survey on a new generation computation paradigm," *Comput. Netw.*, vol. 251, Sep. 2024, Art. no. 110653, doi: 10.1016/j.comnet.2024.110653.
[145] R. Wang, M. A. Kishk, and M.-S. Alouini, "Reliability analysis of multi-hop routing in multi-tier LEO satellite networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 3, pp. 1959–1973, Mar. 2024, doi: 10.1109/TWC.2023.3293467.
[146] Y. Lyu, H. Hu, R. Fan, Z. Liu, J. An, and S. Mao, "Dynamic routing for integrated satellite-terrestrial networks: A constrained multi-agent reinforcement learning approach," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 5, pp. 1204–1218, May 2024.
[147] Z. Xu, W. Quan, N. Cheng, M. Liu, J. Deng, L. Han, and D. Gao, "RP-ER: Relative position based efficient routing mechanism for LEO satellite network," in *Proc. IEEE Global Commun. Conf.*, Dec. 2023, pp. 5967–5972, doi: 10.1109/GLOBECOM54140.2023.10437726.

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